

Chapter II: Prohibited AI Practices

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I. Introduction

Article 5 addresses use cases of artificial intelligence that are deemed **incompatible with European values** (and, by extension, fundamental rights) and thus warrant prohibition as an *ultima ratio* for the legislator.¹ Chapter II is hence primarily grounded in values and

¹ Recital 28 AIA. Article 2 TEU provides that 'the Union is founded on the values of respect for human dignity, freedom, democracy, equality, the rule of law and respect for human rights, including the rights of persons belonging to minorities.'

ethical considerations, reflecting concerns over socially undesirable applications of AI that pose risks to fundamental rights.² This reflects the regulation's overarching objective of establishing the European Union as a region where only trustworthy AI systems and general-purpose AI models are used.³ Chapter II is based on Article 114 of the Treaty of the Functioning of the European Union ('TFEU'), while Article 16 TFEU serves as a legal basis for the specific rules on the processing of personal data in relation to the prohibition on the use of remote biometric identification systems, biometric categorization systems, and individual risk assessments for law enforcement purpose.⁴

- 3.02** The Act's outright and qualified prohibitions have been a **point of contention** since its inception. Debates on this issue can be traced back to discussions within the High-Level Expert Group on AI, particularly regarding whether 'red lines' should be established for AI.⁵ Throughout the legislative process, Article 5 has undergone considerable modification, underscoring its controversial nature. The initial Commission draft only contained versions of the prohibitions that are now to be found in Article 5(1)(a), (b), (c), and (h).⁶ While the Council did not suggest any significant modifications to these prohibitions, the European Parliament pushed for a significant widening of the originally suggested provisions, in addition to additional prohibitions.⁷ The expansion of the list of prohibitions needs to be seen as part of a broader push to emphasize the fundamental rights dimension of the AI Act in the late stages of the legislative procedure.⁸
- 3.03** While Chapter II and Article 5 are titled 'Prohibited AI Practices', the actual **scope** of these prohibitions warrants careful examination. Although Article 5 ostensibly establishes outright prohibitions, most provisions are qualified prohibitions that, upon closer analysis, may not operate as true prohibitions. This implies that while the provision serves a prohibitive function in certain respects, it simultaneously functions as a legitimizing instrument for '**non-prohibited practices**' that may be lawfully deployed under its regulatory framework. This stands in contrast with the objective to devise 'clear red lines' that was laid out in the Impact Assessment.⁹ During the legislative procedure, it was noted time and time again that the AI practices listed in Article 5 are deemed to be of 'unacceptable risk' to health, safety, and fundamental rights.¹⁰ Some of the usages of artificial intelligence governed by

² Recital 28 AIA reads as follows: 'Aside from the many beneficial uses of AI, it can also be misused and provide novel and powerful tools for manipulative, exploitative and social control practices. Such practices are particularly harmful and abusive and should be prohibited because they contradict Union values of respect for human dignity, freedom, equality, democracy and the rule of law and fundamental rights enshrined in the Charter, including the right to nondiscrimination, to data protection and to privacy and the rights of the child.'

³ Article 1(1) AIA.

⁴ Recital 3 AIA. See further Consolidated version of the Treaty on the Functioning of the European Union Protocol (No 21) on the position of the United Kingdom and Ireland in the area of freedom, security and justice (AFSJ) [2016] OJ C202/295 and Protocol (No 22) on the position of Denmark [2012] OJ C326/1.

⁵ Thomas Metzinger, 'EU Guidelines: Ethics Washing Made in Europe' *Der Tagesspiegel Online* (8 April 2019) <<https://www.tagesspiegel.de/politik/ethics-washing-made-in-europe-5937028.html>> accessed 25 June 2025.

⁶ Lines 178–196b of the Proposal for a Regulation of the European Parliament and of the Council laying down harmonized rules on artificial intelligence (Artificial Intelligence Act) and amending certain Union legislative acts (Four Columns AI Act) [2024] COM (2021) 206 final (Final draft as updated on 21 January 2024) <<https://artificialintelligenceact.eu/wp-content/uploads/2024/01/AIA-Final-Draft-21-January-2024.pdf>> accessed 17 March 2025.

⁷ Ibid.

⁸ Marco Almada and Nicolas Petit, 'The EU AI Act: A Medley of Product Safety and Fundamental Rights?' (2023) Robert Schuman Centre for Advanced Studies Research Paper 2023/59 <<https://cadmus.eui.eu/handle/1814/75982>> accessed 14 March 2025.

⁹ Commission, 'Commission staff working document - Impact Assessment - Accompanying the Proposal for a Regulation of the European Parliament and of the Council laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) and amending certain union legislative acts' SWD (2021) 84 final, 50.

¹⁰ Commission, 'AI Act' <<https://digital-strategy.ec.europa.eu/en/factpages/ai-act>> accessed 27 January 2025.

Article 5 are also unlawful under other pre-existing legal frameworks, including the Charter of Fundamental Rights. In such scenarios, the AI Act adds another layer of protection within a broader framework safeguarding fundamental rights. Indeed, Article 5(8) emphasizes that the AI Act does not override prohibitions arising from other Union laws when AI practices infringe upon them.¹¹

It is noteworthy that Article 5 regulates **AI practices**, in contrast to the rest of the regulation, which focuses on AI systems or, in the case of Chapter V, general-purpose AI models. This distinction in terminology is confusing, particularly as the relationship between AI practices and AI systems is not clarified, and the regulation only defines the AI systems but not the AI practices.¹² This concept reflects that the use cases are considered contrary to EU values and pose risks to fundamental rights not primarily because of the use of AI, but rather because the underlying practice itself is deemed problematic. This stands in contrast to the rules on general-purpose AI models, where the technology itself is regarded as inherently risky, regardless of its specific application. Notwithstanding, Article 5 requires the existence of an AI system for the practice to fall within its scope, in line with the regulation's definition of material scope in 1(1) and (2).

There will frequently be **overlaps between the high-risk AI systems** regime in Chapter III and the prohibited practices outlined in Chapter II. Indeed, all 'non-prohibited practices' that benefit from an exemption to a qualified prohibition established by Article 5 qualify as HRAIS.¹³ This interplay highlights the regulatory framework's layered approach, where AI systems may simultaneously fall under different compliance obligations depending on their specific use case and risk profile.¹⁴ Similarly, there are also touching points with **general-purpose AI models** where the AI practice relies on an AI system, which could be a GPAIS or an AI system using a GPAIS without having the capability of serving a variety of purposes.¹⁵ In this case, the obligations for providers of GPAIM outlined in Chapter V also apply to the 'non-prohibited practices' within the scope of Article 5.

Chapter II was the first regime of the AI Act to enter into force on **2 February 2025**.¹⁶ Importantly, it also covers AI systems already placed on the market or put into service prior to this date.¹⁷ This reflects that the AI practices governed by Article 5 are fundamentally incompatible with EU values and pose such serious risks to fundamental rights that they must be prohibited. It should also be noted, however, that national enforcement structures only need to be available by 2 August 2025, meaning that an actual enforcement of the prohibitions prior to this date is unlikely.¹⁸ **National enforcement** could, indeed, emerge as the Achilles heel for Article 5. There is, indeed, a possibility that EU Member States, especially

¹¹ Article 5(8) AIA.

¹² See further Article 3(1) AIA. It is worth noting that the overall objective of the regulation, defined by Article 1(1) is also not limited to AI systems but artificial intelligence in general.

¹³ This implies that the providers of these systems must ensure compliance with relates requirements in Chapter III AIA. See further Articles 8(2) and 16(1) AIA.

¹⁴ Commission, 'Guidelines on prohibited artificial intelligence practices established by Regulation (EU) 2024/1689 (AI Act) C(2025) 884 final, 12.

¹⁵ Articles 5(1) and (2) AIA require the presence of a 'system'. Article 3(66) AIA defines as general-purpose AI system 'means an AI system which is based on a general-purpose AI model and which has the capability to serve a variety of purposes, both for direct use as well as for integration in other AI systems'. See further analysis of Article 3(66) AIA.

¹⁶ Article 113(3)(a) AIA.

¹⁷ Ibid.

¹⁸ Article 113(3)(b) AIA.

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those with poor rule of law standards, could establish legal grounds for after-the-fact remote biometric identification.¹⁹

- 3.07** The legislative provisions in Article 5 have been complemented by **guidelines** from the Commission ('the Commission Guidelines').²⁰ These Guidelines were adopted on 4 February 2025.²¹ In EU law, guidelines are not legally binding.²² Rather, they are a form of soft law that interpret hard law provisions and inform national authorities and individuals about how to apply directives, regulations, or Treaty Articles in practice.²³ Thus, while such guidelines may serve as useful reference points for providers seeking to interpret Article 6(3), they do not confer legal certainty, as their interpretations may ultimately be subject to judicial reversal.
- 3.08** At the request of the Member States or the AIO, or on its own initiative, the Commission can update these guidelines if this is deemed necessary.²⁴
- 3.09** Notably, Article 5 lacks an explicit mechanism for **future changes to Article 5** to address emerging uses and challenges, unlike what is foreseen for HRAIS.²⁵ Article 112(1) provides that the Commission ought to assess the need for amendment of the list of prohibited AI practices once a year following the regulation's entry into force. This obligation ends with the period of the delegation of power laid down in Article 97—namely, on 1 August 2029—unless it is tacitly renewed in line with Article 97(2) or prematurely ended by virtue of Article 97(3). The same procedure applies regarding the review of the list of HRAIS in Annex III.²⁶ A key distinction, however, lies in the fact that these rules are not only addressed but also contextualized and elaborated within the framework of Chapter III.²⁷ This may be seen as an indication that the revision clause was a matter of contention, and that any future attempt by the Commission to invoke this prerogative could prove controversial. An analysis of legislative history confirms that the draft regulation did not anticipate a review of Article 5. Rather, the reference to Article 5 was added in the trilogues following a related suggestion by Parliament.²⁸ In any event, such amendments would have to be addressed by the general revision clause in Article 112(10), which does not empower the Commission to adopt delegated acts but merely to submit amendment proposals.²⁹

II. Article 5: Prohibited AI Practices

1. The following AI practices shall be prohibited:
 - (a) the placing on the market, the putting into service or the use of an AI system that deploys subliminal techniques beyond a person's consciousness or purposefully

¹⁹ Philipp Hacker, 'Comments on the Final Trilogue Version of the AI Act' (*Social Science Research Network*, 23 January 2024) <<https://papers.ssrn.com/abstract=4757603>> accessed 18 March 2025.

²⁰ Article 96(1)(b) AIA. See also Commission Guidelines on prohibited artificial intelligence practices (n 14).

²¹ These guidelines were adopted on the basis of Article 96(1)(b) AIA.

²² Article 288 TFEU.

²³ Oana Stefan, 'European Union Soft Law: New Developments Concerning the Divide Between Legally Binding Force and Legal Effects' (2012) 75 *The Modern Law Review* 879.

²⁴ See further Article 96(2) AIA.

²⁵ See further Article 7 AIA.

²⁶ Article 112(1) AIA.

²⁷ Articles 7(1) and (3) AIA.

²⁸ Article 84(1) dAIA and Four Columns AI Act (n 6) line 760.

²⁹ Contrast with Article 7(1) AIA.

manipulative or deceptive techniques, with the objective, or the effect of materially distorting the behaviour of a person or a group of persons by appreciably impairing their ability to make an informed decision, thereby causing them to take a decision that they would not have otherwise taken in a manner that causes or is reasonably likely to cause that person, another person or group of persons significant harm;

- (b) the placing on the market, the putting into service or the use of an AI system that exploits any of the vulnerabilities of a natural person or a specific group of persons due to their age, disability or a specific social or economic situation, with the objective, or the effect, of materially distorting the behaviour of that person or a person belonging to that group in a manner that causes or is reasonably likely to cause that person or another person significant harm;
- (c) the placing on the market, the putting into service or the use of AI systems for the evaluation or classification of natural persons or groups of persons over a certain period of time based on their social behaviour or known, inferred or predicted personal or personality characteristics, with the social score leading to either or both of the following:
 - (i) detrimental or unfavourable treatment of certain natural persons or groups of persons in social contexts that are unrelated to the contexts in which the data was originally generated or collected;
 - (ii) detrimental or unfavourable treatment of certain natural persons or groups of persons that is unjustified or disproportionate to their social behaviour or its gravity;
- (d) the placing on the market, the putting into service for this specific purpose, or the use of an AI system for making risk assessments of natural persons in order to assess or predict the risk of a natural person committing a criminal offence, based solely on the profiling of a natural person or on assessing their personality traits and characteristics; this prohibition shall not apply to AI systems used to support the human assessment of the involvement of a person in a criminal activity, which is already based on objective and verifiable facts directly linked to a criminal activity;
- (e) the placing on the market, the putting into service for this specific purpose, or the use of AI systems that create or expand facial recognition databases through the untargeted scraping of facial images from the internet or CCTV footage;
- (f) the placing on the market, the putting into service for this specific purpose, or the use of AI systems to infer emotions of a natural person in the areas of workplace and education institutions, except where the use of the AI system is intended to be put in place or into the market for medical or safety reasons;
- (g) the placing on the market, the putting into service for this specific purpose, or the use of biometric categorisation systems that categorise individually natural persons based on their biometric data to deduce or infer their race, political opinions, trade union membership, religious or philosophical beliefs, sex life or sexual orientation; this prohibition does not cover any labelling or filtering of lawfully acquired biometric datasets, such as images, based on biometric data or categorizing of biometric data in the area of law enforcement;
- (h) the use of 'real-time' remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement, unless and in so far as such use is strictly necessary for one of the following objectives:

- (i) the targeted search for specific victims of abduction, trafficking in human beings or sexual exploitation of human beings, as well as the search for missing persons;
 - (ii) the prevention of a specific, substantial and imminent threat to the life or physical safety of natural persons or a genuine and present or genuine and foreseeable threat of a terrorist attack;
 - (iii) the localisation or identification of a person suspected of having committed a criminal offence, for the purpose of conducting a criminal investigation or prosecution or executing a criminal penalty for offences referred to in Annex II and punishable in the Member State concerned by a custodial sentence or a detention order for a maximum period of at least four years. Point (h) of the first subparagraph is without prejudice to Article 9 of Regulation (EU) 2016/679 for the processing of biometric data for purposes other than law enforcement.
2. The use of 'real-time' remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement for any of the objectives referred to in paragraph 1, first subparagraph, point (h), shall be deployed for the purposes set out in that point only to confirm the identity of the specifically targeted individual, and it shall take into account the following elements:
- (a) the nature of the situation giving rise to the possible use, in particular the seriousness, probability and scale of the harm that would be caused if the system were not used;
 - (b) the consequences of the use of the system for the rights and freedoms of all persons concerned, in particular the seriousness, probability and scale of those consequences. In addition, the use of 'real-time' remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement for any of the objectives referred to in paragraph 1, first subparagraph, point (h), of this Article shall comply with necessary and proportionate safeguards and conditions in relation to the use in accordance with the national law authorising the use thereof, in particular as regards the temporal, geographic and personal limitations. The use of the 'real-time' remote biometric identification system in publicly accessible spaces shall be authorised only if the law enforcement authority has completed a fundamental rights impact assessment as provided for in Article 27 and has registered the system in the EU database according to Article 49. However, in duly justified cases of urgency, the use of such systems may be commenced without the registration in the EU database, provided that such registration is completed without undue delay.
3. For the purposes of paragraph 1, first subparagraph, point (h) and paragraph 2, each use for the purposes of law enforcement of a 'real-time' remote biometric identification system in publicly accessible spaces shall be subject to a prior authorisation granted by a judicial authority or an independent administrative authority whose decision is binding of the Member State in which the use is to take place, issued upon a reasoned request and in accordance with the detailed rules of national law referred to in paragraph 5. However, in a duly justified situation of urgency, the use of such system may be commenced without an authorisation provided that such authorisation is requested without undue delay, at the latest within 24 hours. If such authorisation is rejected, the use shall be stopped with immediate effect and all the data, as

well as the results and outputs of that use shall be immediately discarded and deleted. The competent judicial authority or an independent administrative authority whose decision is binding shall grant the authorisation only where it is satisfied, on the basis of objective evidence or clear indications presented to it, that the use of the 'real-time' remote biometric identification system concerned is necessary for, and proportionate to, achieving one of the objectives specified in paragraph 1, first subparagraph, point (h), as identified in the request and, in particular, remains limited to what is strictly necessary concerning the period of time as well as the geographic and personal scope. In deciding on the request, that authority shall take into account the elements referred to in paragraph 2. No decision that produces an adverse legal effect on a person may be taken based solely on the output of the 'real-time' remote biometric identification system.

4. Without prejudice to paragraph 3, each use of a 'real-time' remote biometric identification system in publicly accessible spaces for law enforcement purposes shall be notified to the relevant market surveillance authority and the national data protection authority in accordance with the national rules referred to in paragraph 5. The notification shall, as a minimum, contain the information specified under paragraph 6 and shall not include sensitive operational data.
5. A Member State may decide to provide for the possibility to fully or partially authorise the use of 'real-time' remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement within the limits and under the conditions listed in paragraph 1, first subparagraph, point (h), and paragraphs 2 and 3. Member States concerned shall lay down in their national law the necessary detailed rules for the request, issuance and exercise of, as well as supervision and reporting relating to, the authorisations referred to in paragraph 3. Those rules shall also specify in respect of which of the objectives listed in paragraph 1, first subparagraph, point (h), including which of the criminal offences referred to in point (h)(iii) thereof, the competent authorities may be authorised to use those systems for the purposes of law enforcement. Member States shall notify those rules to the Commission at the latest 30 days following the adoption thereof. Member States may introduce, in accordance with Union law, more restrictive laws on the use of remote biometric identification systems.
6. National market surveillance authorities and the national data protection authorities of Member States that have been notified of the use of 'real-time' remote biometric identification systems in publicly accessible spaces for law enforcement purposes pursuant to paragraph 4 shall submit to the Commission annual reports on such use. For that purpose, the Commission shall provide Member States and national market surveillance and data protection authorities with a template, including information on the number of the decisions taken by competent judicial authorities or an independent administrative authority whose decision is binding upon requests for authorisations in accordance with paragraph 3 and their result.
7. The Commission shall publish annual reports on the use of real-time remote biometric identification systems in publicly accessible spaces for law enforcement purposes, based on aggregated data in Member States on the basis of the annual reports referred to in paragraph 6. Those annual reports shall not include sensitive operational data of the related law enforcement activities.
8. This Article shall not affect the prohibitions that apply where an AI practice infringes other Union law.

III. Article 5(1)(a): Subliminal, Manipulative or Deceptive Techniques

- 3.10** Article 5(1)(a) prohibits the placing on the market, the putting into service or the use of an AI system that deploys subliminal techniques beyond a person's consciousness or purposefully manipulative or deceptive techniques, with the objective, or the effect of materially distorting the behaviour of a person or a group of persons by appreciably impairing their ability to make an informed decision. These AI practices, moreover, need to cause individuals to take a decision that they would not have otherwise taken in a manner that causes or is reasonably likely to cause that person, another person or group of persons significant harm.

A. Overview

- 3.11** Manipulation is a matter of general concern at a time where an increasing amount of human time, attention, and energy is spent online.³⁰ Some of these techniques are now commonly referred to as 'dark (commercial) patterns', which have been defined as practices found in on-line user interfaces that 'steer, deceive, coerce, or manipulate consumers into making choices that often are not in their best interests'.³¹ Articles 5(1)(a) and (b) are designed to address the harms that arise when 'persons engage in unwanted behaviours, or to deceive them by nudging them into decisions in a way that subverts and impairs their autonomy, decision-making and free choices'.³² One may argue that this is precisely what common recommendation algorithms are designed to achieve.³³
- 3.12** Article 5(1)(a) is grounded in the regulation's objective to prevent **risks to fundamental rights** where AI is used to subliminally influence, manipulate, or deceive.³⁴ The Impact Assessment deemed that the fundamental right to human dignity³⁵ is at risk when AI is used for manipulation to 'appeal to the subliminal perception of individuals, thus causing them to take decisions that are beyond their cognitive capacities'.³⁶ Human dignity is threatened if AI systems reduce individuals to mere tools for achieving certain ends.³⁷ Manipulative techniques also have the potential to violate other fundamental rights, such as the right for respect of mental integrity³⁸ or the right to freedom of thought.³⁹ AI can, indeed, be 'misused

³⁰ OECD, 'Dark Commercial Patterns' (2022) OECD Digital Economy Papers 336 <https://www.oecd.org/en/publications/dark-commercial-patterns_44f5e846-en.html> accessed 25 June 2025; Harry Brignull, 'Dark Patterns: Deception vs. Honesty in UI Design' (*A List Apart*, 1 November 2011) <<https://alistapart.com/article/dark-patterns-deception-vs-honesty-in-ui-design/>> accessed 27 January 2025.

³¹ OECD Dark Commercial Patterns (n 30). For an analysis of this concept, see further Mark R Leiser, "Dark Patterns": The Case for Regulatory Pluralism between the European Union's Consumer and Data Protection Regimes' in Eleni Kosta, Ronald Leenes and Irene Kamara (eds), *Research Handbook on EU Data Protection Law* (Edward Elgar Publishing 2022) and Uta Kohl, 'Toxic Recommender Algorithms: Immunities, Liabilities and the Regulated Self-Regulation of the Digital Services Act and the Online Safety Act' (2024) 16 *Journal of Media Law* 301.

³² Recital 29 AIA.

³³ Kohl (n 31).

³⁴ Recital 28 AIA.

³⁵ Article 1 CFR.

³⁶ Commission Impact Assessment (n 9) 17. See also Commission Guidelines on prohibited artificial intelligence practices (n 14) 18.

³⁷ Commission Guidelines on prohibited artificial intelligence practices (n 14) 18.

³⁸ Article 3(1) CFR.

³⁹ Article 10(1) CFR.

and provide novel and powerful tools for manipulative, exploitative and social control practices.⁴⁰

Article 5(1)(a) catches three distinct but connected variants of manipulation, namely subliminal techniques beyond a person's consciousness, purposefully manipulative techniques, and deceptive techniques.⁴¹ This creates terminological complexity as Article 5(1)(a) treats these three techniques as distinct concepts, while Recital 29 collectively categorizes them as 'AI-enabled manipulative techniques'. Consequently, the term 'manipulative techniques' operates both as an umbrella concept encompassing all three techniques within Article 5(1)(a)'s scope and as a designation for one of the specific sub-categories enumerated in that provision.

Regarding **legislative history**, the Commission draft foresaw an outright prohibition of the placing on the market, putting into service or use of an AI system that deploys 'subliminal techniques beyond a person's consciousness in order to materially distort a person's behaviour in a manner that causes or is likely to cause that person or another person physical or psychological harm'.⁴² The Council mandate was largely identical but clarified that the prohibition applied to systems 'with the objective or effect' of material distortion of behaviour and that the resulting harm needed to be 'reasonably likely' rather than just likely.⁴³ The Parliament mandate suggested deeper modifications such as that the prohibition should apply in addition to 'purposefully manipulative or deceptive techniques' with the objective or effect of materially distorting the behavior of a person or group of persons by 'appreciably impairing the person's ability to make an informed decision, thereby causing the person to take a decision that that person would not have otherwise taken'.⁴⁴ It further suggested replacing 'physical or psychological harm' with 'significant harm'.⁴⁵

B. Analysis

Four criteria need to be met for an AI practice to fall within the **scope** of Article 5(1)(a). (i) An AI system needs to be placed on the market, put into service or used, which, (ii) either deploys subliminal techniques beyond a person's consciousness or purposefully manipulative or deceptive techniques that, (iii) materially distorts the behaviour of a person or a group of persons. The distortion must appreciably impair their ability to make an informed decision, resulting in a decision that the person or the group of persons would not have otherwise taken. Finally, (iv) the distorted behaviour must cause or be reasonably likely to cause significant harm to that person, another person, or a group of persons. While this provision appears to establish a broad prohibition of certain AI systems, its multiple conditions—including undefined legal concepts such as 'significant harm'—might substantially narrow its application and reduce its prohibitive effect.

⁴⁰ Recital 28 AIA.

⁴¹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 19 and 23.

⁴² Article 5(1)(a) dAIA and Four Columns AI Act (n 6) line 181.

⁴³ Four Columns AI Act (n 6) line 181.

⁴⁴ Ibid.

⁴⁵ Ibid.

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i. The placing on the market, the putting into service or the use of an AI system

3.16 Article 5(1)(a) prohibits the placing on the market, the putting into service, or the use of an AI system⁴⁶ that deploys subliminal techniques beyond a person's consciousness or purposefully manipulative or deceptive techniques. '**Placing on the market**' refers to the first making available of an AI system or a general-purpose AI model on the Union market.⁴⁷ '**Putting into service**' refers to the supply of an AI system for first use directly to the deployer or for a provider's own use in the EU for its intended purpose.⁴⁸ The natural or legal persons who place AI systems on the market or put them into service qualify as providers, provided that they also developed or ordered the development of the system.⁴⁹ Unlike Articles 5(1)(d)–(g), there is no requirement that the AI system is put into service explicitly for the purpose caught by this provision.

3.17 Article 5(1)(a) also covers the **use** of the AI system. Natural or legal persons using an AI system under their authority legally qualify as deployers.⁵⁰ The concept of use is not legislatively defined. The Commission Guidelines suggest that the concept 'should be understood in a broad manner to cover the use or deployment of the system at any moment of its lifecycle after having been placed on the market or put into service'.⁵¹ It follows that Article 5(1)(a) (and all other provisions in Chapter II with the exception of Article 5(1)(h), which applies to deployers only) is one of the few provisions in the regulation that apply to **deployers and providers** in an otherwise largely provider-centric legal framework.⁵² The Guidelines emphasize that while providers must consider the conditions of use that may be reasonably foreseen prior to the placement on the market of their AI system, deployers are responsible 'for taking the lawful conditions for the use of the system into account'.⁵³

3.18 Importantly, the AI Act does not prohibit the pure **development** of manipulative AI systems in the EU that are marketed internationally.⁵⁴ This rule reflects the regulation's market-oriented approach, allowing activities that may benefit the market, even if they do not fully align with the principles of human-centric and trustworthy AI.⁵⁵ Indeed, the regulation focuses on preventing the 'harmful effects of AI systems *within the Union*'.⁵⁶ This approach mirrors other areas of EU law, such as the regulation of pesticides, which are prohibited for use within the Union but may still be manufactured domestically for export to third countries.⁵⁷

⁴⁶ For a definition, refer to Article 3(1) AIA.

⁴⁷ Article 3(9) AIA. Article 3(10) AIA defines 'making available on the market' as 'means the supply of an AI system or a general-purpose AI model for distribution or use on the Union market in the course of a commercial activity, whether in return for payment or free of charge'.

⁴⁸ Article 3(11) AIA.

⁴⁹ Article 3(3) AIA.

⁵⁰ Article 3(4) AIA. Note that this is not the case where they do so during a personal non-professional activity. See further Article 2(10) AIA.

⁵¹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 5.

⁵² See, e.g. Articles 16, 53 and 55 AIA.

⁵³ Commission Guidelines on prohibited artificial intelligence practices (n 14) 5. Commission Notice, 'The Blue Guide' on the implementation of EU products rules 2022' [2022] OJ C247/1, section 2.8.

⁵⁴ See also Articles 2(1) and (8) AIA.

⁵⁵ See further Article 1(1) AIA.

⁵⁶ Article 1(1) AIA (my own emphasis).

⁵⁷ Silke Bollmohr and Susan Haffmans, 'Imports and Exports: Banned but Sold Anyway' (*Heinrich Böll Stiftung*, 18 October 2022) <<https://eu.boell.org/en/PesticideAtlas-imports-exports>> accessed 9 December 2024.

ii. Subliminal techniques beyond a person's consciousness

Article 5(1)(a) firstly catches subliminal techniques beyond a person's consciousness. These are techniques that deploy subliminal components, such as audio, image, video stimuli that are beyond human perception.⁵⁸ Because subliminal techniques are inherently beyond consciousness and hence covert, they 'bypass a person's rational defences against manipulation and are capable of influencing decisions without the conscious awareness of the person, raising significant ethical concerns and impairing individual autonomy, agency and free choice.'⁵⁹ Subliminal techniques must operate beyond consciousness, meaning that they must be capable of influencing behaviour in ways in which the person remains **unaware** of how the technique works or of its effects on their decision-making or value and opinion formation.⁶⁰ These techniques can include all 'stimuli delivered through audio, visual, or tactile media that are too brief or subtle to be noticed'.⁶¹ Subliminal influence can also materialize through machine-brain interfaces or virtual reality 'as they allow for a higher degree of control of what stimuli are presented to persons'.⁶² In Commission briefings, examples given of such manipulation were 'an inaudible sound played in truck drivers' cabin to push them to drive longer than healthy and safe' and dolls that have an 'integrated voice assistant which encourages a minor to engage in progressively dangerous behavior or challenges in the guise of a fun or cool game'.⁶³ Mark Leiser has noted that the formulation beyond consciousness implies that the manipulation must be undetectable by the conscious mind and potent enough to skew behaviour materially.⁶⁴ This posits a dual challenge, 'the first is the demarcation of the threshold where influence recedes from the conscious arena into the shadowy depths of the subconscious, and the second is the evidentiary burden to establish that such subliminal manoeuvring is causally linked to discernible harm'.⁶⁵

The Commission Guidelines provide **further examples** of practices deemed to be caught by Article 5(1)(a). First, visual subliminal messages, where an AI system incorporates images or text into video playback that, while technically perceptible, appear so briefly that they evade conscious recognition but still subconsciously shape attitudes or influence behaviour.⁶⁶ Second, auditory subliminal messages, where an AI system uses auditory stimuli by embedding sounds or verbal messages at low volumes or camouflaging them within other audio, allowing them to influence the listener without their conscious awareness.⁶⁷ Other examples cited include subvisual and subaudible cueing and embedded images.⁶⁸

⁵⁸ Recital 29 AIA.

⁵⁹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 20. These explanations rely on contested philosophical concepts of free choice and free will, whose very existence remains subject to ongoing debate. See further Timothy O'Connor and Christopher Franklin, 'Free Will' in Edward N Zalta and Uri Nodelman (eds), *The Stanford Encyclopedia of Philosophy* (Metaphysics Research Lab, Stanford University 2022).

⁶⁰ Recital 29 AIA. Commission Guidelines on prohibited artificial intelligence practices (n 14) 20.

⁶¹ Ibid.

⁶² Recital 29 AIA.

⁶³ DG CONNECT, 'ELLIS Workshop in Human-Centric Machine Learning' (10 May 2021) presentation by Gabriele Mazzini, 'A European Strategy for Artificial Intelligence' 2:52:26–2:54 <<https://www.youtube.com/watch?v=OZtuVKWqhl0&t=10346s>> cited in Michael Veale and Frederik Zuiderveen Borgesius, 'Demystifying the Draft EU Artificial Intelligence Act — Analysing the Good, the Bad, and the Unclear Elements of the Proposed Approach' (2021) 22 Computer Law Review International 97, 99.

⁶⁴ Leiser (n 31) 17.

⁶⁵ Ibid.

⁶⁶ Commission Guidelines on prohibited artificial intelligence practices (n 14) 20.

⁶⁷ Ibid.

⁶⁸ Ibid. It is unclear why this is different from visual cues.

3.21 Article 5(1)(a) is, *inter alia*, designed to be **forward-looking**, anticipating future technological advancements and their potential risks for fundamental rights such as developments in ‘emerging machine-brain interfaces and advanced techniques like dream-hacking and brain spyware’.⁶⁹ Indeed, it is worth questioning whether many contemporary techniques would meet the high thresholds established by Article 5(1)(a), and whether the co-legislators genuinely intended this provision to apply to current forms of manipulation, or rather to more speculative, science-fiction-like future scenarios.⁷⁰ In this context, it must be noted that subliminal influence is already **prohibited by other legal frameworks**.⁷¹ The Council of Europe Convention on Transfrontier Television prohibits subliminal techniques in advertising and tele-shopping.⁷² Similarly, the Audiovisual Media Services Directive prohibits subliminal techniques in audiovisual commercial communications.⁷³ The Impact Assessment deemed that subliminal techniques could still threaten human dignity despite these sector-specific norms.⁷⁴

iii. Purposeful manipulation

3.22 Article 5(1)(a) further targets purposefully manipulative techniques. This prohibition covers techniques that subvert or impair a person’s autonomy, decision-making or free choice in ways that people are not consciously aware of, or, where they are aware of, can still be deceived or unable to control or resist.⁷⁵ The Commission Guidelines argue for an interpretation of the concept as ‘techniques that are designed or objectively aim to influence, alter, or control an individual’s behaviour in a manner that undermines their individual autonomy and free choices’.⁷⁶ Manipulative techniques are typically designed to exploit cognitive biases, psychological vulnerabilities, or situational factors that make individuals more susceptible to influence.⁷⁷ Thus, in contrast to subliminal techniques, which are by definition covert, manipulative need not necessarily be. An example of a manipulative technique is personalized manipulation where an AI system that creates and tailors highly persuasive messages based on an individual’s personal data or exploits other

⁶⁹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 21 noting that ‘the prohibition in Article 5(1)(a) AI Act targets only cases of such significantly harmful subliminal manipulation and not machine-brain interface applications in general when designed in a safe and secure manner and respectful of privacy and individual autonomy’.

⁷⁰ Michael Veale and Frederik Zuiderveen Borgesius, ‘Demystifying the Draft EU Artificial Intelligence Act — Analysing the Good, the Bad, and the Unclear Elements of the Proposed Approach’ (2021) 22 *Computer Law Review International* 97, 98–99.

⁷¹ For a discussion from the perspective of ethics, see Juan Pablo Bermúdez and others, ‘What Is a Subliminal Technique? An Ethical Perspective on AI-Driven Influence’, 2023 *IEEE International Symposium on Ethics in Engineering, Science, and Technology (ETHICS)* (IEEE 2023) <<https://ieeexplore.ieee.org/document/10155039/>> accessed 26 June 2025.

⁷² Article 13(2) of the Council of Europe - European Convention on Transfrontier Television (opened for signature 17 April 1989, entered into force 1 May 1993) ETS No 132 <<https://www.coe.int/en/web/conventions/-/council-of-europe-european-convention-on-transfrontier-television-ets-no-132-translations>> accessed 4 February 2025.

⁷³ Article 9(1)(b) of the Directive 2010/13/EU of the European Parliament and of the Council of 10 March 2010 on the coordination of certain provisions laid down by law, regulation or administrative action in Member States concerning the provision of audiovisual media services (Audiovisual Media Services Directive) [2010] OJ L95/1; On the harms of subliminal manipulation in advertising, see Bahador Bahrami, Nilli Lavie and Geraint Rees, ‘Attentional Load Modulates Responses of Human Primary Visual Cortex to Invisible Stimuli’ (2007) 17 *Current Biology* 509.

⁷⁴ Commission Impact Assessment (n 9) 17.

⁷⁵ Recital 29 AIA.

⁷⁶ These definitions again focus on free choice and free will, the very existence of which remains contested in philosophy. See further O’Connor and Franklin (n 59).

⁷⁷ Commission Guidelines on prohibited artificial intelligence practices (n 14) 21.

individual vulnerabilities, thereby influencing their behaviour or choices to a point of creating significant harm.⁷⁸

Article 5(1)(a) explicitly requires that manipulation must be **purposeful**—meaning that it must have been deliberately pursued by the provider or deployer. While the manipulation itself has to be purposeful, ‘it is not necessary that the provider or deployer or the system itself deploying the manipulative techniques also intends to cause harm.’⁷⁹ The Commission Guidelines imply that this purpose need not originate from a human agent, but that it can also emanate from the AI system, meaning that Article 5(1)(a) also catches AI systems that ‘manipulate individuals without any human intending them to do so.’⁸⁰ This could be the case where, regardless of whether the provider intends it, it is the AI system itself that deploys such manipulative techniques, rather than the provider or the deployer who has designed or used the system in this way.⁸¹

There can be **overlaps** between subliminal techniques and purposefully manipulative techniques where the manipulative technique operates beyond the threshold of conscious awareness.⁸² Finally, it should be noted that manipulation is also addressed by other norms, such as Article 5 UCPD,⁸³ which prohibits unfair commercial practices.⁸⁴ These norms can apply cumulatively to the same practice.⁸⁵

iv. Deceptive techniques

Article 5(1)(a) also applies to **deceptive techniques**. These are techniques that undermine or weaken a person’s autonomy, decision-making, or free choice in ways that the individual may not be consciously aware of, or, if aware, may still be deceived or unable to control or resist.⁸⁶ According to the Commission Guidelines, deceptive techniques should be understood ‘to involve presenting false or misleading information with the objective or the effect of deceiving individuals and influencing their behaviour in a manner that undermines their autonomy, decision-making and free choices.’⁸⁷ An example of a deceptive technique is the use of an AI chatbot that impersonates a relative with synthetic voice and tries to pretend it is the person causing scams and significant harms.⁸⁸ The Guidelines make the case that the concept of deceptive techniques includes AI systems that deceive individuals without any human intent to do so, such as ‘AI systems may learn deceptive techniques simply because this increases their performance for the task for which they were developed, for example by reinforced learning.’⁸⁹

⁷⁸ Ibid.

⁷⁹ Ibid. On the legal status of recitals in EU law, see further related analysis in the Introduction

⁸⁰ Commission Guidelines on prohibited artificial intelligence practices (n 14) 21.

⁸¹ Ibid 22.

⁸² Ibid 21. (‘While not all manipulative techniques operate beyond the threshold of conscious awareness, many do and there may be an overlap with subliminal techniques, since such techniques also ultimately have manipulative effects.’)

⁸³ UCPD.

⁸⁴ Catalina Goanta, ‘Regulatory Siblings: The Unfair Commercial Practices Directive Roots of the AI Act’ in Inge Graef and Bart Van Der Sloot (eds), *The Legal Consistency of Technology Regulation in Europe* (1st edn, Hart Publishing 2024) 71.

⁸⁵ Article 2(9) AIA and Commission Guidelines on prohibited artificial intelligence practices (n 14) 24.

⁸⁶ Recital 29 AIA.

⁸⁷ Commission Guidelines on prohibited artificial intelligence practices (n 14) 22.

⁸⁸ Ibid 23.

⁸⁹ Ibid.

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- 3.26** An important question in relation to deceptive techniques is whether the term ‘**purposefully**’ in the text of Article 5(1)(a) relates only to manipulative or also to deceptive techniques.⁹⁰ The Guidelines appear to implicitly suggest that deceptive techniques need not be purposeful.⁹¹
- 3.27** The concept of deceptive techniques relates closely to **deepfakes**, which are regulated by Chapter IV of the regulation.⁹² The related obligations in Articles 50(1), (2), and (4) serve as a mitigating measure that helps reduce the risk of deception that may occur once AI-generated content is made public, and lessens the potential for harmful distortions in an individual’s opinion formation, beliefs, and behaviour.⁹³
- 3.28** Deceptive practices are also caught by Article 25(1) DSA, which prohibits the design, organization or operation of ‘online interfaces in a way that deceives or manipulates service recipients or which otherwise materially distorts or impairs the ability of the service recipients to make free and informed decisions.’⁹⁴ The provision is, however, unable to catch all deceptive design practices as ‘any subterfuge of design that resides beneath the superficial layer eludes the strictures of Article 25.’⁹⁵ These practices include sensory manipulation, algorithmic manipulation, behavioural conditioning, system dependency, and control and structural manipulation.⁹⁶
- 3.29** All three techniques listed in Article 5(1)(a) can apply **cumulatively**. Regarding the overlap of purposefully manipulative and deceptive techniques, the Commission Guidelines note that where both are applied in combination, this may significantly influence the behaviour of individuals, leading them to make decisions based on unconscious manipulations and false beliefs. This can create a ‘feedback loop where individuals are less likely to question or critically evaluate the information received, since the manipulative elements have already primed their cognitive biases and emotional responses.’⁹⁷
- v. The objective, or the effect of materially distorting the behaviour of a person or a group of persons
- 3.30** For a subliminal, purposefully manipulative, or deceptive technique to fall within the scope of Article 5(1)(a), it must have the objective, or the effect of materially distorting the behaviour of a person or a group of persons. This implies that the technique must have a ‘substantial impact on the behaviour where a person’s autonomy and free choices are undermined, rather than a minor influence.’⁹⁸ The Commission Guidelines argue that intent is not a necessary requirement given that Article 5(1)(a) also covers practices that only have the ‘effect’ of causing material distortion.⁹⁹ Yet, there should be ‘a plausible/reasonably likely causal link between the potential material distortion of the behaviour and the subliminal, purposefully manipulative or deceptive technique deployed by the AI system.’¹⁰⁰ This formulation could lead to complex interpretational issues, as the difference between

⁹⁰ As noted above, even with respect to manipulation, the degree of intentionality required remains uncertain.

⁹¹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 22–23.

⁹² For a definition, see Article 3(60) AIA.

⁹³ Commission Guidelines on prohibited artificial intelligence practices (n 14) 23.

⁹⁴ For an analysis of the DSA see Martin Husovec, *Principles of the Digital Services Act* (1st edn, OUP 2024).

⁹⁵ Leiser (n 31) 9.

⁹⁶ Ibid 10–15.

⁹⁷ Commission Guidelines on prohibited artificial intelligence practices (n 14) 24.

⁹⁸ Ibid.

⁹⁹ Ibid.

¹⁰⁰ Ibid.

a plausible link and a reasonably likely causal link may be difficult to discern in practice. Ultimately, this connection can only be evaluated *ex post*. In practice, proving that an action was solely caused by such influence will be extremely challenging without access to counterfactual information.

Article 5(1)(a) requires that there be a **material distortion of the behaviour** of a person or group of persons. According to the Commission Guidelines, the notion ‘appreciably impairing’ should be interpreted as referring to a ‘substantially reduced ability to make informed and autonomous decisions, thereby causing individuals to behave in a way or to take a decision that they would otherwise not have taken. It goes beyond minor or negligible impacts and involves a significant distortion or hindrance in decision-making and free choice, including in relation to opinion- and belief-formation.’¹⁰¹ An appreciable impairment hence involves a degree of coercion, manipulation or deception.¹⁰² The Commission Guidelines clarify that an objective to materially distort behaviour may be present where the objective is pursued by the provider, the deployer, or ‘the system itself within the implicit objectives it may pursue’.¹⁰³ The AI Act does not define what constitutes a **material distortion of behaviour**. However, the Guidelines clarify that it occurs when there is a ‘substantial impact, rather than a minor or trivial one’.¹⁰⁴ According to Recital 29, an intention to distort behaviour may not exist where ‘the distortion results from factors external to the AI system which are outside the control of the provider or the deployer, namely factors that may not be reasonably foreseeable and therefore not possible for the provider or the deployer of the AI system to mitigate’.¹⁰⁵

The notion of the **informed decision** is very context-specific and requires ‘an understanding and knowledge of the relevant information, including the available options, the risks and benefits of each choice, the possible effects of the AI system on their behaviour, and, as appropriate, other contextual information that is important for the decision-making or the behaviour of the person’.¹⁰⁶ This concept has to be interpreted in alignment with established interpretations in EU consumer protection law. Article 2(e) UCPD provides that ‘to materially distort the economic behaviour of consumers’ should be interpreted as ‘using a commercial practice to appreciably impair the consumer’s ability to make an informed decision, thereby causing the consumer to take a transactional decision that he would not have taken otherwise’.¹⁰⁷ In order to assess this criterion, courts ought to adopt the perspective of the average consumer’s decision-making process.¹⁰⁸ There is no need to prove that a consumer’s economic behaviour has been distorted, it suffices to establish that a commercial practice is ‘likely’ (i.e. capable) of impacting an average consumer’s transactional decision.¹⁰⁹ Indeed, even accurate information may be misleading if presented in a way that distorts the consumer’s decision-making process.¹¹⁰

¹⁰¹ Ibid 25.

¹⁰² Ibid.

¹⁰³ Ibid 26.

¹⁰⁴ Ibid 38.

¹⁰⁵ Recital 29 AIA.

¹⁰⁶ Commission Guidelines on prohibited artificial intelligence practices (n 14) 25.

¹⁰⁷ Article 2(e) UCPD.

¹⁰⁸ Recitals 18 and 19 UCPD.

¹⁰⁹ Case C-611/14 *Canal Digital Danmark* EU:C:2016:800, para 7. See further Commission Guidelines on prohibited artificial intelligence practices (n 14) 25.

¹¹⁰ See Case C-281/12 *Trento Sviluppo und Centrale Adriatica* EU:C:2013:859 and Commission Guidelines on prohibited artificial intelligence practices (n 14) 25.

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vi. Causes or is reasonably likely to cause significant harm

3.33 Finally, an AI practice caught by the above criteria only falls within the scope of Article 5(1)(a) if it causes or is reasonably likely to cause significant harm.¹¹¹ Article 5(1)(a) covers a closed list of harms, for which Recital 29 uses the more open-ended formulation of ‘significant harms, in particular having sufficiently important adverse impacts on physical, psychological health or financial interests.’¹¹² The Commission Guidelines make the case that Article 5(1)(a) should not be interpreted as proving a closed list of harms in stating that the ‘main types of harms [...] include physical, psychological, financial, and economic harms that may be compound with broader societal harms in certain cases.’¹¹³ This is justified by the fact that Recital 28 explains that the prohibitions can also cause broader societal harms and contradict Union values of respect for human dignity, freedom, equality, democracy, the rule of law, and fundamental rights enshrined in the Charter, as well as by the fact that Article 1(1) mentions the EU values of democracy and the rule of law.¹¹⁴ Considering the broad and open-ended nature of these concepts and the fact that while this interpretation is read into the recital but absent from Article 5(1)(a), one can anticipate that judges will, in the future, need to clarify the boundaries of the concept of harm in Article 5(1)(a).

3.34 The Commission Guidelines provide interpretations of the various kinds of harm that are explicitly covered in the legislative text. **Physical harm** is interpreted as encompassing ‘any injury or damage to a person’s life, health and material damage to property.’¹¹⁵ **Psychological harm** refers to an exploitation of cognitive and emotional vulnerabilities and influence an individual’s behaviours in ways that can cause significant harm.¹¹⁶ These harms not only include adverse effects on mental health, which is narrowly defined, but also ‘psychological and emotional well-being’, which are of course very broad formulations.¹¹⁷ This emphasizes that diagnosing a psychological pathology is not required for an effect to be considered psychological harm. According to the Guidelines, such harms are especially concerning because they can accumulate gradually and may not be immediately visible, yet they can lead to long-term and serious consequences.¹¹⁸ However, measuring them is more challenging, requiring a case-by-case evaluation to assess their severity, considering all relevant circumstances.¹¹⁹ **Financial and economic harm** is interpreted as encompassing ‘a range of adverse effects, including financial loss, financial exclusion, economic instability’ such as a chatbot that offers fraudulent products that lead to ‘significant’ financial harm.¹²⁰ Importantly, these effects can manifest themselves in combination, leading to compounded and multifaceted negative impact.¹²¹

¹¹¹ Article 5(1)(a) AIA and Recital 29 AIA.

¹¹² Recital 29 AIA. This has been criticized, with some suggesting that societal harm should have been included, as manipulation can lead to broader negative impacts on society. Risto Uuk, ‘Manipulation and the AI Act’ (*future of life institute*, 18 January 2022) <https://futureoflife.org/wp-content/uploads/2022/08/FLI-Manipulation_AI_Act.pdf> accessed 2 February 2025 (‘harming the democratic process, eroding the rule of law, or exacerbating inequality, can cause only modest harms to individuals but hurt societies at large.’)

¹¹³ Commission Guidelines on prohibited artificial intelligence practices (n 14) 28.

¹¹⁴ Ibid footnote 76.

¹¹⁵ Ibid 28.

¹¹⁶ Ibid.

¹¹⁷ Ibid.

¹¹⁸ Ibid 29.

¹¹⁹ Ibid.

¹²⁰ Ibid.

¹²¹ Ibid.

Recital 29 clarifies that ‘it is not necessary for the provider or the deployer to have the intention to cause significant harm, provided that such harm results from the manipulative or exploitative AI-enabled practices.’¹²² The Commission Guidelines underline the context-specific nature of the harm requirement as the individual effects ‘should always be material and significant in each case.’¹²³ The Guidelines furthermore devise a list of key considerations that can be derived from other areas of EU law and could be taken into account when assessing this criterion, namely the severity of harm, the context of harm and possible cumulative effects, the scale and intensity of harm, the affected person’s vulnerability, as well as the duration and possible reversibility of the harm.¹²⁴

The concept ‘**reasonably likely**’ serves to determine whether there is a ‘plausible/reasonably likely causal link between the manipulative or deceptive technique’ and the harm.¹²⁵ This implies ‘a judgement of reasonableness on an objective basis and according to universally accepted criteria (e.g. technical and scientific), including a criterion of rationality in establishing plausible causality between the AI practice and the significant harm that may arise.’¹²⁶ The Guidelines stress that opacity or transparency of the AI system and its functioning may affect the conclusion regarding this causal link and, hence, the application of the prohibition.¹²⁷ This is an important caveat, as it highlights that the most complex AI systems may may fall outside the scope of this prohibition.

The harm requirement also serves to further define the scope of Article 5(1)(a) by excluding techniques beyond a person’s consciousness that are not constitutive of harm. These are for instance subliminal manipulation techniques, which aid motivation and could be self-consciously chosen by the user with their full and informed consent¹²⁸ and which do not cause them harm. Notwithstanding, there is considerable judicial discretion in defining the thresholds for harm, which will significantly impact the scope of the regulation.

vii. Exemptions

Recital 29 formulates two exemptions to Article 5(1)(a).¹²⁹ It firstly provides that the prohibition should not affect **lawful medical treatment practices**, including the psychological treatment of a mental disease or physical rehabilitation carried out in accordance with the applicable law and medical standards, for example explicit consent of the individuals or their legal representatives.¹³⁰ It is questionable whether this exception, placed in a recital and therefore not strictly legally binding, was necessary, given that such practices cannot typically be considered inherently harmful.¹³¹

Second, Recital 29 also provides that **common and legitimate commercial practices, for example in the field of advertising**, that comply with the applicable law should not, in themselves, be regarded as constituting harmful manipulative AI-enabled practices. This

¹²² Recital 29 AIA.

¹²³ Commission Guidelines on prohibited artificial intelligence practices (n 14) 30.

¹²⁴ Ibid 30–31.

¹²⁵ Ibid 31.

¹²⁶ Ibid.

¹²⁷ Ibid.

¹²⁸ Nathalie A Smuha and others, ‘How the EU Can Achieve Legally Trustworthy AI: A Response to the European Commission’s Proposal for an Artificial Intelligence Act’ (*Social Science Research Network*, 5 August 2021) 22 <<https://papers.ssrn.com/abstract=3899991>> accessed 18 March 2025.

¹²⁹ On the legal status of recitals in EU law, see further related analysis in the Introduction.

¹³⁰ Recital 29 AIA.

¹³¹ On the legal status of recitals in EU law, see further related analysis in the Introduction.

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exception is much more ambiguous and raises questions about the interpretation of 'legitimate', which differs from the concept of 'lawful' as applied to medical practices. Similar uncertainties arise regarding the meaning of 'commercial practices', a notion that is in any event very wide. It hence appears that in the commercial domain, which arguably covers most AI practices that aim to manipulate, Article 5(1)(a) only applies where the practice is also caught by other legal frameworks, such as where they constitute a violation of data protection, consumer protection, or fundamental rights law. This exception, hidden away in a recital, is a very notable limitation of scope in an age of toxic recommendation algorithms¹³², which has been characterized by a 'journey from the overt manipulation of user interfaces to the covert shaping of user experiences by algorithms signifies an escalating complexity in consumer protection and regulation.'¹³³ Depending on the legal weight the CJEU assigns to this recital, it could substantially undermine the practical effect of the prohibition laid down in Article 5(1)(a).

IV. Article 5(1)(b): Harmful Exploitation of Individual or Group Vulnerabilities

- 3.40** Article 5(1)(b) prohibits the placing on the market, the putting into service, or the use of an AI system that exploits any of the vulnerabilities of a natural person or a specific group of persons due to a closed list of three grounds, namely their age, a disability or a specific social or economic situation. To fall into the scope of this prohibition, this exploitation of individual or group vulnerabilities needs to have the objective, or the effect, of materially distorting the behaviour of that person or a person belonging to that group in a manner that causes or is reasonably likely to cause that person or another person significant harm.

A. Overview

- 3.41** Article 5(1)(b) formulates an outright prohibition regarding the exploitation of individual or group vulnerabilities based on three grounds, namely age, disability, or a specific social or an economic situation. This prohibition echoes concerns that artificial intelligence can be a tool used to exploit vulnerabilities.¹³⁴
- 3.42** The prohibitions in Article 5(1)(a) and (b) can apply cumulatively.¹³⁵ The key distinctions between them are twofold: First, Article 5(1)(b) requires a lower threshold of influence, as the exploitation of vulnerability does not need to rise to the level of manipulation. Second, Article 5(1)(b) applies only to a narrowly defined closed list of vulnerabilities. The main factor aiding the determination of whether Article 5(1)(a) and (b) apply is the determination

¹³² Kohl (n 31).

¹³³ Leiser (n 31) 7.

¹³⁴ See further Chenyue Wang and others, 'The Artificial Intelligence Divide: Who Is the Most Vulnerable?' (2024) New Media & Society; On vulnerability more generally, see Gianclaudio Malgieri, *Vulnerability and Data Protection Law* (1st edn, OUP 2023) and Malou Beck and Sofia Ranchordás, 'Vulnerability' in Mareile Kaufmann and Heidi Mork Lomell (eds), *Handbook of digital criminology* (De Gruyter 2024).

¹³⁵ Commission Guidelines on prohibited artificial intelligence practices (n 14) 18 (stating that the prohibitions in Articles 5(1)(a) and 5(1)(b) 'aim to prevent manipulation and exploitation through AI systems that reduce individuals to mere tools for achieving certain ends and to safeguard those that are most vulnerable and susceptible to harmful manipulation and exploitation.'): 43 (anticipates the cumulative application of both prohibitions).

of whether the exploitation applies regardless of the specific vulnerabilities of the persons concerned or not.¹³⁶

Regarding **legislative history**, the original Commission draft suggested banning AI systems that exploit vulnerabilities related to age, physical or mental ability to materially distort their behaviour in a manner likely to cause that or another person harm.¹³⁷ The European Parliament suggested that only ‘significant’ harm should be caught by the prohibition but also pushed for a widening of its scope to include vulnerabilities related to personality traits, social or economic situation. It also specified that the vulnerabilities could be either those of a ‘person or a specific group of persons’ and that systems with both ‘the objective or the effect of’ distorting behaviour should be caught.¹³⁸ The Council suggested that the system needed to be ‘reasonably’ likely to cause physical or psychological harm to be caught by the provision.¹³⁹

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B. Analysis

Article 5(1)(b) applies if four cumulative conditions are met. First, the practice must constitute the ‘placing on the market’, the ‘putting into service’, or the ‘use’ of an AI system. This criterion has been examined in detail in relation to Article 5(1)(a) and the reader is referred to this analysis. Second, the AI system must exploit vulnerabilities due to age, disability, or socio-economic situations. Third, this exploitation of vulnerability must have the objective, or the effect of materially distorting the behaviour of a person or a group of persons. Finally, the distorted behaviour must cause or be reasonably likely to cause significant harm to that person, another person, or a group of persons. Article 5(1)(b) hence requires that there be a plausible causal link between the exploitation, the material distortion of the behaviour of the person, and the significant harm that has resulted, or is reasonably likely to result, from that behaviour.¹⁴⁰

3.44

i. Closed list of three vulnerabilities

Article 5(1)(b) covers a closed list of three vulnerabilities, namely age, disability or a specific social or economic situation. These vulnerabilities are deemed to make individuals ‘particularly susceptible to manipulative and exploitative practices.’¹⁴¹ This approach has been criticized as it does not include characteristics such as sex or race, which are also protected characteristics under EU equality law.¹⁴² This omission has been attributed to the idea that characteristics such as sex or race do not result in a reduction of capacities.¹⁴³ The European approach is also in line with the UNESCO Recommendation on the Ethics of Artificial Intelligence (2021) which emphasizes inclusivity and fairness in AI development and deployment, and calls for special attention to vulnerable groups, including children, older

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¹³⁶ Ibid 43.

¹³⁷ Article 5(1)(b) dAIA and Four Columns AI Act (n 6) line 182.

¹³⁸ Ibid.

¹³⁹ Ibid.

¹⁴⁰ Commission Guidelines on prohibited artificial intelligence practices (n 14) 33.

¹⁴¹ Ibid 34.

¹⁴² Article 21 CFR. Smuha and others (n 128) 22.

¹⁴³ Rostam J Neuwirth, ‘Prohibited Artificial Intelligence Practices in the Proposed EU Artificial Intelligence Act (AIA)’ (2023) 48 Computer Law & Security Review 105798.

people, and people with disabilities.¹⁴⁴ The Council of Europe Framework Convention on Artificial Intelligence embraces a narrower approach by obliging contracting parties to ‘take due account of any specific needs and vulnerabilities in relation to respect for the rights of persons with disabilities and of children.’¹⁴⁵

- 3.46** The notion of ‘**vulnerability**’ is not legislatively defined.¹⁴⁶ The Commission Guidelines underline that vulnerabilities ‘encompass a broad spectrum of categories, including cognitive, emotional, physical, and other forms of susceptibility that can affect the ability of an individual or a group of persons to make informed decisions or otherwise influence their behaviour.’¹⁴⁷ However, the protective scope of Article 5(1)(b) is limited to age, disability and socio-economic situations as the individuals concerned are deemed to ‘in principle have more limited capacity to recognise or resist the AI manipulative or exploitative practices and are in need of enhanced protection.’¹⁴⁸
- 3.47** Age, disability and socio-economic situations are also not explicitly defined in the regulation. **Age** can be presumed to include both young and old age. In fact, Recital 28 mentions children as a vulnerable group and serves as a reminder that the rights of the child are protected in EU law.¹⁴⁹ Article 24(1) of the Charter posits that **children** have the right to such protection and care as is necessary for their well-being.¹⁵⁰ This reference to well-being is also of relevance in debates around digital well-being considering that children are more prone to digital overuse and addiction.¹⁵¹ During the legislative process, the European Commission gave the following example of an exploitation of vulnerability due to young age—a doll with an integrated voice assistant that encourages a minor to engage in progressively dangerous behaviour or challenges in the guise of a cool game.¹⁵² The Commission Guidelines confirm that Article 5(1)(b) seeks to protect against the AI-based exploitation of ‘cognitive and other limitations that children [...] may have, and to protect them from harmful undue influence, manipulation and exploitation.’¹⁵³ Children are defined as persons below the age of 18.¹⁵⁴
- 3.48** At the same time, Recital 58 recognizes a general vulnerability associated with **old age** and research has confirmed that AI can generate risks for seniors, for instance, cybersecurity risks associated with generative AI.¹⁵⁵ The Guidelines suggest that Article 5(1)(b) seeks to protect against the AI-based exploitation of ‘cognitive and other limitations that [...] older people may have, and to protect them from harmful undue influence, manipulation and exploitation.’¹⁵⁶ While the Commission deems that children are always vulnerable, older

¹⁴⁴ UNESCO, ‘Recommendation on the Ethics of Artificial Intelligence’ (2021) <<https://unesdoc.unesco.org/ark:/48223/pf0000380455>> accessed 26 June 2025. See further Commission Guidelines on prohibited artificial intelligence practices (n 14) 34.

¹⁴⁵ Article 18 of Council of Europe Framework Convention on Artificial Intelligence and Human Rights, Democracy and the Rule of Law (opened for signature 5 September 2024, not yet in force) CETs No 225.

¹⁴⁶ Compare with Article 5(3) UCPD.

¹⁴⁷ Commission Guidelines on prohibited artificial intelligence practices (n 14) 34.

¹⁴⁸ Ibid.

¹⁴⁹ Article 24 CFR.

¹⁵⁰ Article 24(1) CFR.

¹⁵¹ Simin Cao and Hui Li, ‘A Scoping Review of Digital Well-Being in Early Childhood: Definitions, Measurements, Contributors, and Interventions’ (2023) 20 International Journal of Environmental Research and Public Health 3510.

¹⁵² Mazzini (n 63).

¹⁵³ Commission Guidelines on prohibited artificial intelligence practices (n 14) 34.

¹⁵⁴ Ibid.

¹⁵⁵ Zahm Siyed, ‘Generative AI Increases Cybersecurity Risks for Seniors’ (2024) 17 Computer and Information Science 1.

¹⁵⁶ Commission Guidelines on prohibited artificial intelligence practices (n 14) 34.

people ‘*might suffer* from reduced cognitive capacities (even if not suffering from dementia) and *might struggle* with the complexities of modern AI technologies, making them in those cases vulnerable to scams or coercive tactics.’¹⁵⁷ This raises the question of whether a generalized protection based solely on old age is warranted, given that cognitive abilities do not uniformly decline with age, nor do they do so in a linear or consistent manner across the population. Furthermore, while young age is clearly defined as under 18 years of age, there is no precise threshold for when old age-related vulnerability begins. Provocatively, one might ask whether a 55-year-old could claim protection under Article 5(1)(b). It is worth emphasizing that the prohibition only covers exploitations of vulnerabilities and does not stand in the way of use cases of artificial intelligence designed to benefit seniors and children.¹⁵⁸

Disability has to be interpreted in accordance with the European Accessibility Act, which defines ‘persons with disabilities’ as ‘persons who have long-term physical, mental, intellectual or sensory impairments which in interaction with various barriers may hinder their full and effective participation in society on an equal basis with others.’¹⁵⁹ According to the Commission Guidelines, AI applications that are not designed in an accessible manner should not automatically be regarded to exploit vulnerabilities of persons with disabilities since they do not specifically target those vulnerabilities, but are simply inaccessible.¹⁶⁰ This raises challenging questions about the threshold for determining when an AI system is considered to merely be designed in an inaccessible manner versus being designed to discriminate through inaccessibility. Article 5(1)(b) only covers exploitations of vulnerabilities and does not stand in the way of use cases of artificial intelligence designed to benefit persons with disabilities.¹⁶¹

The reference to the **specific social or economic situation** of individuals or groups is the vaguest ground covered by Article 5(1)(b). Recital 29 provides a non-exhaustive list of specific social or economic situations, which are those ‘likely to make those persons more vulnerable to exploitation such as persons living in extreme poverty, ethnic or religious minorities.’¹⁶² The reference to extreme poverty raises the question of whether a restrictive interpretation of Article 5(1)(b) is warranted. Indeed, extreme poverty has been defined as living on less than US\$3.00 per person per day at 2021 purchasing power parity—a notably high threshold that significantly narrows the range of economic situations falling within the scope of this provision.¹⁶³ Considering the much narrower formulation in Article 5(1)(b), a broader interpretation is warranted. Indeed, the Commission Guidelines recommend that the list in Recital 29 should be treated as non-exhaustive and that economic situations such as temporary unemployment and over-indebtedness should also be covered.¹⁶⁴

¹⁵⁷ Ibid 35 my own emphasis).

¹⁵⁸ See further Martha E Pollack, ‘Intelligent Technology for an Aging Population: The Use of AI to Assist Elders with Cognitive Impairment’ (2005) 26 AI Magazine 9; David Mhlanga, ‘Artificial Intelligence in Elderly Care: Navigating Ethical and Responsible AI Adoption for Seniors’ in David Mhlanga and Mufaro Dzingirai (eds), *Fostering Long-Term Sustainable Development in Africa: Overcoming Poverty, Inequality, and Unemployment* (Springer Nature Switzerland 2024).

¹⁵⁹ Recital 29 AIA and Article 3(1) of European Accessibility Act (Directive (EU) 2019/882 of the European Parliament and of the Council of 17 April 2019 on the accessibility requirements for products and services [2019] OJ L151/70).

¹⁶⁰ Commission Guidelines on prohibited artificial intelligence practices (n 14) 36.

¹⁶¹ Gemma Lewin and others, ‘Nature and Prevalence of Long-Term Conditions in People With Intellectual Disability, a Study That Combines the Powers of AI, Big Data and Lived Experience’ (2024) 10 Psych Open S55.

¹⁶² Recital 29 AIA.

¹⁶³ This definition is embraced by the UN. See ‘Goal 1: End Poverty in All Its Forms Everywhere’ (*United Nations Sustainable Development Goals*) <<https://www.un.org/sustainabledevelopment/poverty/>> accessed 9 October 2025.

¹⁶⁴ Commission Guidelines on prohibited artificial intelligence practices (n 14) 37.

- 3.51** According to the Commission Guidelines, persons in disadvantaged socio-economic situations are usually more vulnerable and have fewer resources and lower digital literacy than the general population, which makes it harder for them to discern or counteract exploitative AI practices.¹⁶⁵ Yet, it can be questioned whether financial status or belongingness to an ethnic or religious minority necessarily results in lower digital literacy. Consider the example of a highly educated single parent facing economic hardship. While they are in a vulnerable financial position, their digital literacy may exceed that of the average citizen, raising questions about the broad classification of vulnerability.
- 3.52** The reference to a ‘**specific**’ social-economic situation should be construed not as a unique personal characteristic ‘but rather a legal status or membership to a specific vulnerable social or economic group.’¹⁶⁶ This could be interpreted to include other categories, such as the working poor, which do not have a legal status or membership indicative of their socio-economic situation yet might be just as vulnerable to harm compared to the unemployed.¹⁶⁷
- ii. The objective or the effect of materially distorting the behaviour of that person or a person belonging to that group
- 3.53** Article 5(1)(b) only prohibits practices that exploit individual or group vulnerabilities with the objective or the effect of materially distorting the behaviour of that person or a person belonging to that group.¹⁶⁸
- 3.54** The AI Act does not define what constitutes a **material distortion of behaviour**. However, the Commission Guidelines clarify that it occurs when there is a ‘substantial impact, rather than a minor or trivial one.’¹⁶⁹ The material distortion of behaviour can be intentional or unintentional. According to Recital 29, an intention to distort behaviour may not exist where ‘the distortion results from factors external to the AI system which are outside the control of the provider or the deployer, namely factors that may not be reasonably foreseeable and therefore not possible for the provider or the deployer of the AI system to mitigate.’¹⁷⁰ Importantly, the wording of Article 5(1)(b) implies that this does not require a material change in the behaviour of the group as a whole, but only in the behaviour of an individual within the group.
- iii. Cause or is reasonably likely to cause that person or another person significant harm
- 3.55** Article 5(1)(b) prohibits practices that exploit individual or group vulnerabilities that cause or are reasonably likely to cause that person or another person significant harm. Significant harm can be interpreted as a sufficiently important adverse impact on physical or psychological health, or financial interests that is likely to occur and is particularly dangerous.¹⁷¹ Recital 29 clarifies that harm does not need to arise from a single event but can also encompass harm accumulated over time.¹⁷² What matters is the existence of harm. It is not necessary for the provider or deployer to have an intention to cause such harm, provided that the harm results from the manipulative or exploitative AI-enabled practices.¹⁷³ The

¹⁶⁵ Ibid 36.

¹⁶⁶ Ibid 37.

¹⁶⁷ Luca Ratti and Paul Schoukens (eds), *Working yet Poor: Challenges to EU Social Citizenship* (1st edn, Hart Publishing 2023).

¹⁶⁸ Article 5(1)(b) AIA.

¹⁶⁹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 38.

¹⁷⁰ Recital 29 AIA.

¹⁷¹ Ibid.

¹⁷² Ibid.

¹⁷³ Ibid.

Commission Guidelines recommend a broad interpretation of harm considering that harms can be especially severe and multifaceted due to the heightened susceptibility of vulnerable groups to exploitation.¹⁷⁴ Risks considered tolerable for adults may amount to disproportionate harm when experienced by children or other vulnerable individuals. Accordingly, where uncertainty exists and the potential for serious impact cannot be ruled out, a precautionary stance becomes imperative.¹⁷⁵

The Commission Guidelines provide the following example of an AI system exploitative of the vulnerabilities of children: 'A game uses AI to analyse children's individual behaviour and preferences based on which it creates personalised and unpredictable rewards through addictive reinforcement schedules and dopamine-like loops to encourage excessive play and compulsive usage. The game is designed to be highly addictive, exploiting the vulnerabilities inherent to children, including their limited ability to understand long-term consequences, susceptibility to pressure, lack of self-control, and inclination towards instant gratification. The consequences of this AI-enabled exploitation can be severe and long-lasting for children, including potentially addi[c]tive behaviour, physical health problems due to lack of exercise and sleep, deteriorated eyesight, problems with concentration and reduced cognitive capacities, poor academic performance, and social difficulties. It can significantly impact a child's development and well-being, with potential longer-term consequences that may also extend into adulthood.'¹⁷⁶ This example hints at the future difficulties in enforcing this norm. Indeed, some would argue that all games present these characteristics and harms.¹⁷⁷ The prediction of 'severe and long-lasting effects' is, moreover, difficult, as it requires a projection into the future, which can only be speculative.

This creates a clear tension with the requirement to establish a causal link between the act of exploitation, the material distortion of the individual's behaviour, and the significant harm that has resulted or is reasonably likely to result.¹⁷⁸ Indeed, it raises a fundamental question—how can causality be demonstrated between a present intervention and a future event whose occurrence remains uncertain? Another plausible interpretation of the requirement for long-lasting effects on children, in conjunction with Article 5(1)(b)'s standard that significant harm be 'reasonably likely to result', is that the focus lies on whether the behaviour in question typically tends to lead to harm. However, this interpretation poses its own challenges, as it demands generalized evidence of harm that is often scarce, inconclusive, or contradictory in the context of emerging AI systems.¹⁷⁹

The Commission Guidelines draw a black-and-white picture of AI systems, stating that the regulation targets 'exploitation and addiction-like practices that seriously harm children' but not 'AI-enabled toys, games, learning applications or other digital applications in general that can bring benefits and are not affected if they do not fulfil all conditions for that prohibition'.¹⁸⁰ This binary view of AI systems is an oversimplification that overlooks

¹⁷⁴ Commission Guidelines on prohibited artificial intelligence practices (n 14) 39.

¹⁷⁵ Ibid.

¹⁷⁶ Ibid 34.

¹⁷⁷ Jonathan Haidt, *The Anxious Generation: How the Great Rewiring of Childhood Is Causing an Epidemic of Mental Illness* (Penguin 2024).

¹⁷⁸ Commission Guidelines on prohibited artificial intelligence practices (n 14) 35.

¹⁷⁹ 'Screen Time and Emotional Problems in Kids: A Vicious Circle?' (*American Psychological Association*, 9 June 2025) <<https://www.apa.org/news/press/releases/2025/06/screen-time-problems-children>> accessed 26 June 2025.

¹⁸⁰ Commission Guidelines on prohibited artificial intelligence practices (n 14) 35.

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a crucial issue in digital consumption and related policy-making—the same product can simultaneously serve as a valuable tool or an addictive mechanism. Indeed, even a learning application can be purposefully designed to be highly addictive.

- 3.59** While Article 5(1)(a) addresses **group harms**, Article 5(1)(b) does not. However, Recital 29, which applies to both provisions, references group harms.¹⁸¹ The Commission Guidelines suggest that the concept of group harm should be considered when interpreting Article 5(1)(b) to ensure both prohibitions are applied consistently and in line with the ‘safety logic’ of the regulation as well as the ‘objective of the prohibition in Article 5(1)(b) to protect all individuals belonging to the specific vulnerable groups due to age, disability and specific socio-economic situation’.¹⁸² In practice, this means that the assessment of the significance of harm must take into consideration that harms can be externalized and affect other persons, even those not directly affected by the AI system.¹⁸³ It remains uncertain whether the CJEU would endorse such an interpretation, as it effectively elevates the legal weight of the recital beyond its formal status as an aid to interpretation rather than a binding provision.¹⁸⁴

V. Article 5(1)(c): Social Scoring

- 3.60** Article 5(1)(c) formulates an outright prohibition concerning the placing on the market, the putting into service, or the use of an AI system for the evaluation or classification of natural persons or groups of persons over a certain period of time based on their social behaviour or known, inferred, or predicted personal or personality characteristics, with the social score leading to two outcomes, which can, but do not have to arise cumulatively. First, detrimental or unfavourable treatment of certain natural persons or groups of persons in social contexts that are unrelated to the contexts in which the data was originally generated or collected. Second, detrimental or unfavourable treatment of certain natural persons or groups of persons that is unjustified or disproportionate to their social behaviour or its gravity.

A. Overview

- 3.61** Significant controversy surrounded social scoring systems during the AI Act negotiations, particularly because similar systems began emerging in various jurisdictions around the world, in particular in China.¹⁸⁵ However, as the European Commission itself noted, some EU Member States had also already started implementing social scoring systems.¹⁸⁶ Notwithstanding, these systems are deemed to be contrary to EU values and to pose risks

¹⁸¹ Recital 29 AIA. On the legal status of recitals in EU law, see further related analysis in the Introduction.

¹⁸² Commission Guidelines on prohibited artificial intelligence practices (n 14) 41.

¹⁸³ Ibid. This relates to the discussion of group harms in fundamental right generally, which are deemed to not be sufficiently protected by the individualistic fundamental rights protected in EU law and Western legal orders generally. See instead of many Jef Ausloos, Jill Toh and Alexandra Giannopoulou, ‘How the GDPR Can Exacerbate Power Asymmetries and Collective Data Harms’ (*Ada Lovelace Institute*, 29 November 2022) <<https://www.adalovelaceinstitute.org/blog/gdpr-power-asymmetries-collective-data-harms/>> accessed 26 June 2025.

¹⁸⁴ On the legal status of recitals in EU law, see further related analysis in the Introduction.

¹⁸⁵ Commission Impact Assessment (n 9) 17.

¹⁸⁶ Ibid 47, footnote 222 notes that ‘systems where people are socially scored with discriminatory or disproportionately adverse treatment have also been put in place in some Member States such as the Gladaxe system in Denmark or the SyRI system in the Netherlands’.

to fundamental rights.¹⁸⁷ Social scoring can, indeed, lead to discriminatory outcomes, the marginalization of certain groups, as well as infringements of the rights to dignity and non-discrimination.¹⁸⁸ Social scoring can, moreover, undermine the values of equality and justice.¹⁸⁹ The ban on certain social scoring practices is justified by the fact that ‘mass scale citizens’ scoring would unduly restrict individuals’ fundamental rights and be contrary to the values of democracy, freedom and the principles that all people should be treated as equals before the law and with dignity.’¹⁹⁰ Article 5(1)(c) hence forms one of the measures devised to protect against the harmful effects AI systems can have on fundamental rights.¹⁹¹

Regarding **legislative history**, the HLEGAI had already recommended a prohibition of social scoring.¹⁹² The Commission draft would have prohibited the placing on the market, putting into service or use of AI systems ‘by public authorities or on their behalf for the evaluation or classification of the trustworthiness of natural persons’ used ‘over a certain period of time based on their social behaviour or known or predicted personality characteristics’ and where the social score leads to either or both: (a) detrimental treatment of an individual or group in contexts unrelated to those where the data was originally collected; or (b) detrimental treatment of an individual or group that is unjustified or disproportionate to their social behaviour or its gravity.¹⁹³ Limiting the scope of the prohibition to public actors only was deemed problematic as private actors can also have access to enormous amounts of personal data, covering many important areas of life.¹⁹⁴ In response to these criticisms, the Parliament and Council proposed expanding the provision’s scope in two significant ways: making it applicable to both public and private actors, and broadening it beyond mere ‘trustworthiness’ evaluations to encompass all forms of evaluation and classification. Furthermore, the Parliament also suggested that the classifications of groups, not just natural persons, should be caught and that the personal or personality characteristics should not just include those that are ‘predicted’ but also those that are ‘inferred’.¹⁹⁵

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B. Analysis

Three cumulative conditions need to be met for an AI system to fall within the scope of Article 5(1)(c). First, the practice must constitute the ‘placing on the market’, the ‘putting into service’, or the ‘use’ of an AI system. This criterion was already examined in depth in

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¹⁸⁷ Recital 31 AIA. Article 2 TEU provides that ‘the Union is founded on the values of respect for human dignity, freedom, democracy, equality, the rule of law and respect for human rights, including the rights of persons belonging to minorities. These values are common to the Member States in a society in which pluralism, non-discrimination, tolerance, justice, solidarity and equality between women and men prevail’.

¹⁸⁸ Recital 31 AIA.

¹⁸⁹ Ibid.

¹⁹⁰ Commission Impact Assessment (n 9) 47.

¹⁹¹ See further Article 1(1) AIA.

¹⁹² High-Level Expert Group on AI, ‘Policy and Investment Recommendations for Trustworthy AI’ (European Commission 2019) <<https://digital-strategy.ec.europa.eu/en/library/policy-and-investment-recommendations-trustworthy-artificial-intelligence>> accessed 4 February 2025.

¹⁹³ See Article 5(1)(c) dAIA and Four Columns AI Act (n 6) line 183.

¹⁹⁴ EDPB-EDPS, ‘Joint Opinion 5/2021 on the Proposal for a Regulation of the European Parliament and of the Council Laying down Harmonised Rules on Artificial Intelligence (Artificial Intelligence Act)’ (18 June 2021) 2 <https://www.edpb.europa.eu/our-work-tools/our-documents/edpb-edps-joint-opinion/edpb-edps-joint-opinion-52021-proposal_en> accessed 9 December 2024 (‘private companies, such as social media and cloud service providers, also can process vast amounts of personal data and conduct social scoring. Consequently, the future AI Regulation should prohibit any type of social scoring’).

¹⁹⁵ Article 5(1)(c) dAIA and Four Columns AI Act (n 6) line 183.

relation to Article 5(1)(a), and the reader is referred to that analysis. Second, the AI system must be intended or used for the evaluation or classification of natural persons or groups of persons over a certain period of time based on: (a) their social behaviour; or (b) known, inferred, or predicted personal or personality characteristics. Third, the social score created with the assistance of the AI system must lead or be capable of leading to the detrimental or unfavourable treatment of persons or groups in one or more of the following scenarios: (a) in social contexts unrelated to those in which the data was originally generated or collected; and/or (b) treatment that is unjustified or disproportionate to their social behaviour or its gravity.¹⁹⁶

i. Evaluation or classification of natural persons or groups of persons

- 3.64** Article 5(1)(c) prohibits the placing on the market, putting into service and use of an AI system, for the evaluation or classification of natural persons or groups of persons. The prohibition applies to the public and private sector.¹⁹⁷ It encompasses systems that **evaluate or classify** natural persons or groups thereof based on multiple data points related to their social behaviour in multiple contexts or known, inferred, or predicted personal or personality characteristics over certain periods of time.¹⁹⁸ The notions of evaluation and classification are not legislatively defined. According to the Commission Guidelines, a classification can simply be based on characteristics, such as age, sex, and height, but evaluation implies an assessment or judgement, which will often also amount to profiling in the sense of Article 3(52), as well as Article 4(4) GDPR and Article 11 LED.¹⁹⁹ As a consequence, these legal regimes may apply cumulatively.²⁰⁰ Indeed, most if not all practices falling within the scope of Article 5(1)(c) are also subject to the GDPR.²⁰¹
- 3.65** Prohibited social scoring practices must rely on **multiple data points** related to social behaviour in multiple contexts or known, inferred, or predicted personal or personality points, yet the regulation leaves this term undefined—offering no clarity on whether, say, eight data points are sufficient.²⁰² That said, in practice, systems operating on such minimal data are unlikely to involve AI, making the issue largely theoretical. These data points must be collected over ‘**certain periods of time**’, confirming that social scoring involves a comprehensive analysis of behavioural patterns rather than isolated or momentary events.²⁰³ This excludes scores based on isolated incidents, instead requiring the identification of behavioural patterns across time.²⁰⁴
- 3.66** Social scoring systems refer exclusively to systems that undertake an evaluation or classification of **natural persons**, and, at least in principle, not legal persons.²⁰⁵ The evaluation and

¹⁹⁶ See further Commission Guidelines on prohibited artificial intelligence practices (n 14) 51.

¹⁹⁷ Recital 31 AIA.

¹⁹⁸ Ibid.

¹⁹⁹ Article 3(52) AIA, Article 4(4) GDPR and Article 11 LED.

²⁰⁰ Article 2(7) AIA.

²⁰¹ It is worth recalling that the GDPR applies irrespective of whether the profiling is done through an AI system or not.

²⁰² Recital 31 AIA.

²⁰³ Ibid.

²⁰⁴ Article 5(1)(c) AIA.

²⁰⁵ Commission Guidelines on prohibited artificial intelligence practices (n 14) 52. The Commission Guidelines however recommend that ‘if legal entities have been evaluated based on an overall score that aggregates the evaluation or classification of a group of natural persons based on their social behaviour or personal or personality characteristics and this score directly affects those persons (e.g., all employees in a company, students in a specific school whose behaviour has been evaluated), the practice may fall within the scope of Article 5(1)(c) AI Act if all other conditions are fulfilled.’ See *ibid* 61.

classification can also relate to groups of persons. The evaluation or classification, moreover, must lead to **scores** based on their social behaviour or personal and personality characteristics, the score generated by the system may take various forms, such as a numerical value (e.g. from 0 to 1), a ranking, or a label.²⁰⁶

ii. Based on social behaviour or known, inferred or predicted personal or personality characteristics

Evaluation and classification practices must be based on **social behaviour** or known, inferred, or predicted personal or personality characteristics to be caught by Article 5(1)(c). This implies that the data points leading to the social score must pertain to social behaviour, explicitly excluding factors like private financial behaviour.²⁰⁷ Similarly excluded from the prohibition are scoring systems that are based on data regarding professional or academic performance to the extent that they do not account for social behaviour. According to the Commission Guidelines, this ‘could include behaviour of individuals and groups of individuals in social and private contexts, such as participation in cultural events, volunteering, etc., but also social behaviour in business contexts, for example the payment of debts, behaviour when using certain services, as well as relations with public and private entities, government, police, and the law (for example, whether a person obeys traffic rules)’.²⁰⁸ It does not matter which entity collects the data that leads to the assessment.²⁰⁹

The scoring can also be based on **known, inferred, or predicted personal or personality characteristics**, introducing significant ambiguity since the provision does not specify acceptable methods for making such predictions. The Commission Guidelines note that this ‘may include a variety of information relating to a person, for example sex, sexual orientation or sexual characteristics, gender, gender identity, race, ethnicity, family situation, address, income, household members, profession, employment or other legal status, performance at work, economic situation, financial liquidity, health, personal preferences, interests, reliability, behaviour, location or movement, level of debt, type of car etc.’²¹⁰

The Commission Guidelines interpret the notions of **known characteristics** as information that is provided to an AI system and is ‘in most cases verifiable information’.²¹¹ In turn, **inferred characteristics** are those based on information that has been inferred from other information (usually by an AI system).²¹² This notion ought to be interpreted in line with the law on profiling under the GDPR, which also makes use of these concepts.²¹³ **Predicted characteristics** are those estimated on patterns with less than complete accuracy.²¹⁴

iii. Two potential outcomes

For a social scoring system to come within the scope of Article 5(1)(c), the score must lead to two potential outcomes and there must be a causal link between the social score and the outcome. These outcomes may arise cumulatively, but cumulative occurrence is not a

²⁰⁶ Ibid 52.

²⁰⁷ Articles 5(1)(c)(i) and (ii) AIA.

²⁰⁸ Commission Guidelines on prohibited artificial intelligence practices (n 14) 54.

²⁰⁹ Ibid 52.

²¹⁰ Ibid 54.

²¹¹ Ibid.

²¹² Ibid.

²¹³ Ibid. Article 29 Data Protection Working Party, ‘Guidelines on Automated individual decision making and Profiling for the purposes of Regulation 2016/679’ (WP251rev.01, 6 February 2018) 7.

²¹⁴ Commission Guidelines on prohibited artificial intelligence practices (n 14) 54.

prerequisite for the prohibition to apply. In either case, it must be demonstrated that the social scoring led to detrimental or unfavourable treatment.²¹⁵ Establishing both the detrimental or unfavourable character of the treatment and a clear causal connection to the social scoring system presents significant evidentiary challenges in practice.

- 3.71** The first potential outcome falling within the scope of Article 5(1)(c) is the detrimental or unfavourable treatment of certain natural persons or groups of persons in social contexts unrelated to the contexts in which the data was originally generated or collected. The treatment resulting from social scoring must, therefore, be adverse in nature.²¹⁶ Article 5(1)(c) does not apply where the outcome of the score is beneficial—such as yielding financial advantages. Unfavourable treatment means ‘that as a result from the scoring, the person or group of persons must be treated less favourably compared to others without necessarily requiring a particular harm or damage.’²¹⁷ In contrast, detrimental treatment requires ‘the person or group of persons to suffer certain harm and detriment from the treatment.’²¹⁸
- 3.72** Article 5(1)(c) is triggered only when a score derived in one context is repurposed in a different, unrelated context. This contextual transfer is key, as it excludes practices like using criminal records to assess the risk of future criminal conduct, which remain confined to the same domain. Finally, the treatment needs to occur in social contexts, a notion that is not legislatively defined. This social behaviour in any event needs to stem from ‘multiple contexts.’²¹⁹ While a causal link between the social score and the treatment is in principle required, the Commission Guidelines recommend that ‘a plausible causal link may also exist in cases where the harmful consequences have not yet materialised but the AI system is intended to or capable of producing such an adverse outcome.’²²⁰ Furthermore, the social score need not be the sole cause of the detrimental or unfavourable treatment.²²¹
- 3.73** The second outcome falling within the scope of Article 5(1)(c) is that the score results in detrimental or unfavourable treatment of certain natural persons or groups of persons that is unjustified or disproportionate to their social behaviour or its gravity. This serves as a fall-back option for usages of social scoring that evade the scope of the prior scenario, specifically where the score and result relate to the same context, but the unfavourable treatment of natural persons or groups of persons is unjustified or disproportionate to the social behaviour or its gravity. Unfavourable treatment means ‘that as a result from the scoring, the person or group of persons must be treated less favourably compared to others without necessarily requiring a particular harm or damage.’²²² In contrast, detrimental treatment requires ‘the person or group of persons to suffer certain harm and detriment from the treatment.’²²³ Such treatment may also fall within the scope of EU anti-discrimination law, in which case, both legal regimes apply cumulatively.
- 3.74** The Commission Guidelines recommend that these notions be interpreted with consideration of the impact and interference on fundamental rights on a case-by-case basis to

²¹⁵ Recital 31 AIA.

²¹⁶ Ibid.

²¹⁷ Commission Guidelines on prohibited artificial intelligence practices (n 14) 56.

²¹⁸ Ibid.

²¹⁹ Recital 31 AIA.

²²⁰ Commission Guidelines on prohibited artificial intelligence practices (n 14) 55.

²²¹ Ibid.

²²² Ibid 56.

²²³ Ibid.

determine ‘whether such treatment is disproportionate for the legitimate aim pursued, taking into account the general principle of proportionality’²²⁴ Beyond these purely legal dimension, general ethical considerations and principles for fairness and social justice should also be accounted for.²²⁵ Furthermore, the treatment needs to be ‘unjustified’, meaning that it lacks a legitimate aim, which is of course a very subjective notion.²²⁶ The **burden of proof** to demonstrate that an AI practice is legitimate and justified lies with the providers and deployers, who must present evidence to market surveillance authorities during checks. This includes, among other things, providing transparency and information about the system’s functioning and the data used.²²⁷

iv. Exceptions

Recital 31 clarifies that the prohibition of social scoring does not apply to **lawful evaluation practices** of individuals carried out for specific purposes in accordance with EU and national laws.²²⁸ Indeed, certain forms of social scoring are already practised within the European Union under regulated conditions.²²⁹ These social scoring practices are subject to other legal constraints, such as those arising under data protection law.²³⁰ According to the Commission Guidelines, this exemption covers use cases where laws specify the types of data relevant for the specific evaluation purposes and ensure that any resulting detrimental or unfavourable treatment of persons is justified and proportionate.²³¹ This exception underlines that Member States have considerable autonomy to carry out social scoring, a practice explicitly deemed contrary to EU values and fundamental rights, provided that they legislatively enable such practices. The interpretation of this exception, which has been tucked away in a recital and may well be considered *contra legem* to Article 5(1)(c) is likely to lead to future debates.²³²

3.75

VI. Article 5(1)(d): Individual Risk-Assessments and Prediction in the Criminal Domain

Article 5(1)(d) formulates an outright prohibition regarding the placing on the market, the putting into service for this specific purpose, or the use of an AI system for making risk assessments of natural persons to assess or predict the risk of a natural person committing a criminal offence. This assessment must be based solely on the profiling of a natural person or on assessing their personality traits and characteristics. This prohibition does not apply to AI systems used to support the human assessment of the involvement of a person in a criminal activity, which is already based on objective and verifiable facts directly linked to a criminal activity.

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²²⁴ Ibid 58.

²²⁵ Ibid.

²²⁶ Ibid.

²²⁷ Ibid 60.

²²⁸ On the legal status of recitals in EU law, see further related analysis in the Introduction.

²²⁹ For an overview, see Anja Geller, ‘Social Scoring durch Staaten: Legitimität nach europäischem Recht - mit Verweisen auf China’ (PhD thesis, Ludwig-Maximilians-Universität München 2022) <<https://edoc.ub.uni-muenchen.de/id/eprint/31151>> accessed 3 February 2025 and Daithí Mac Síthigh and Mathias Siems, ‘The Chinese Social Credit System: A Model for Other Countries?’ (2019) 82 The Modern Law Review 1034.

²³⁰ Case C-634/21 *SCHUFA* EU:C:2023:957.

²³¹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 50.

²³² On the legal status of recitals in EU law, see further related analysis in the Introduction.

A. Overview

- 3.77** Article 5(1)(d) aims to protect fundamental rights from the risks posed by AI-based risk assessment tools in the criminal domain, in particular the presumption of innocence, which is protected by Article 48 of the Charter.²³³ Indeed, in the years prior to the genesis of the AI Act, there was ample controversy about a risk assessment tool from the United States, COMPAS, which was widely used in the criminal justice system, and faced criticism for biases that led to discriminatory outcomes.²³⁴ This raised global awareness about the risks of using artificial intelligence in law enforcement.²³⁵ Article 5(1)(d) was absent from the draft version of the regulation and was added to the final text following a related suggestion in the Parliament mandate.²³⁶

B. Analysis

- 3.78** Three cumulative conditions need to be fulfilled for an AI system to be caught by the prohibition in Article 5(1)(d). First, the practice must constitute the ‘placing on the market’, ‘the putting into service for this specific purpose’ or the ‘use’ of an AI system. This criterion was examined in depth in relation to Article 5(1)(a) and the reader is referred to that analysis. It must, however, be noted that Article 5(1)(d) only governs AI systems that are put into service for the *specific purpose* of individual risk assessment and prediction of criminal offences.²³⁷ This forms an important contrast to the prohibitions in Articles 5(1)(a)–(c), which apply to AI systems put into service without a specific purpose. Neither the regulation nor the Commission Guidelines elaborate on the appropriate interpretation of this additional criterion. It can be reasonably assumed that what is required is the intent to pursue the specific objective of using an AI system to assess or predict the risk of a natural person committing a criminal offence. This intent may undoubtedly originate with the provider, who, as defined by the AI Act, is the natural or legal person putting the system into service.²³⁸ However, it is worth wondering whether such intent could also stem from the deployer, especially given that ‘putting into service’ encompasses the initial supply of the AI system for direct use by the deployer.²³⁹ This raises important interpretive questions about the attribution of purpose within the provider–deployer relationship that must be assessed on a case-by-case basis.
- 3.79** Second, the AI system must make risk assessments that assess or predict the risk of a natural person committing a criminal offence. Third, the risk assessment or the prediction must be based solely on either, or both, of the following: (a) the profiling of a natural person, (b) assessing a natural person’s personality traits and characteristics.²⁴⁰ Systems that fall short

²³³ See also Recital 42 AIA.

²³⁴ Jeff Larson and others, ‘How We Analyzed the COMPAS Recidivism Algorithm’ (*ProPublica*) <<https://www.propublica.org/article/how-we-analyzed-the-compas-recidivism-algorithm>> accessed 4 February 2025.

²³⁵ EUROPOL, ‘AI and Policing: The Benefits and Challenges of Artificial Intelligence for Law Enforcement’ (Publications Office 2023) Europol Innovation Lab observatory report.

²³⁶ Four Columns AI Act (n 6) 189a.

²³⁷ Article 5(1)(d) AIA.

²³⁸ Article 3(3) AIA.

²³⁹ Article 3(11) AIA.

²⁴⁰ Article 5(1)(d) AIA and Commission Guidelines on prohibited artificial intelligence practices (n 14) 65.

of meeting the criteria espoused by Article 5(1)(d) may, however, qualify as high-risk AI systems.²⁴¹

i. Assessing the risk or predicting the likelihood of a person committing a criminal offence

According to the Commission Guidelines, the **risk assessments** subject to Article 5(1)(d) include ‘a variety of advanced AI technologies and analytical methods applied to large amount of often historical data (including socio-economic data, but also police records, etc.) which, in combination with criminology theories, are used to forecast crime as a basis to inform police and law enforcement strategies and action to combat, control, and prevent crime.’²⁴² The limitation to criminal offences also implies that the prohibition does not apply to administrative offences, the prosecution of which is deemed less intrusive to fundamental rights and freedoms.²⁴³ Recital 42 refers to risk assessments as tools that determine the likelihood of an offence. This underlines that this prohibition, at least in principle, covers **future-orientated** usages of these technologies.²⁴⁴ The prohibition should also not be understood as a general prohibition of crime prediction and risk assessment practices, but rather only applies to AI systems making such assessments, in line with the general material scope of the regulation.²⁴⁵ In principle, Article 5(1)(d) applies only to **natural persons** but there may be borderline cases where a natural person acts via a legal entity as a ‘sole trader’ or as an independent professional (e.g. a lawyer). In such circumstances, Article 5(1)(d) may apply provided that all conditions are fulfilled.²⁴⁶

Neither Article 5(1)(d), the accompanying recital, nor the Commission’s guidelines provide an explanation of the notion of a **criminal offence**. This can be contrasted with Article 5(1)(h)(iii), in relation to which the co-legislators determined that Council Framework Decision 2002/584/JHA serves as a basis for the interpretation of the notion of the criminal offence.²⁴⁷ The contrast between this approach and the absence of definition in Article 5(1)(d) can be interpreted as a deference to domestic law.

It should furthermore be noted that this provision is not expressly addressed at law enforcement authorities. Indeed, it can apply to both **public and private actors** as ‘otherwise the prohibition might be easily circumvented, which would call into question its effectiveness.’²⁴⁸ This would, for instance, be the case when private actors are entrusted by law to exercise public authority and public powers for the prevention, investigation, detection, or prosecution of criminal offences, or for the execution of criminal penalties.²⁴⁹ The Commission Guidelines in any event require that a private actor has been entrusted with the related task by a public authority, meaning that it excludes ‘private entities’ ordinary course of business and with the aim of protecting their own private interests, whilst the fact that those risk

²⁴¹ Article 6(2) and Point 6(d) of Annex III AIA.

²⁴² Commission Guidelines on prohibited artificial intelligence practices (n 14) 66.

²⁴³ Ibid.

²⁴⁴ Ibid 66. (stating that these tools are ‘forward-looking and concern future criminal offences (not yet committed) or crimes that are assessed as a risk of being committed at the moment, including in cases of an attempt or preparatory activities undertaken to commit a criminal offence.’)

²⁴⁵ Ibid 67.

²⁴⁶ Ibid 75.

²⁴⁷ Recital 33 and Annex II AIA.

²⁴⁸ Commission Guidelines on prohibited artificial intelligence practices (n 14) 71.

²⁴⁹ Ibid. This reference to criminal penalties does not directly flow from the legislative text.

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assessments may relate to the risk of criminal offences being committed merely as a purely accidental and secondary circumstance.²⁵⁰ Importantly, the prohibition may also overlap with activities by private entities assessing or predicting the risk of a person of committing a crime where this is objectively necessary for compliance with a legal obligation to which that private operator is subject to assess or predict the risk of persons committing specific criminal offences, such as anti-money laundering and terrorist financing checks.²⁵¹ This also hints at the blurry lines between law enforcement activities caught by Article 2(1)(d) and activities related to national security, which fall outside the scope of the regulation.²⁵² According to the Commission Guidelines, Article 5(1)(d) only applies in the presence of an explicit goal of detecting crime that has been entrusted to the entity by law enforcement authorities.²⁵³ This thus excludes situations where risks assessments occur in the context of private entities' ordinary course of business and with the aim of protecting their own private interests, and the fact that those risk assessments may relate to the risk of criminal offences is purely accidental and secondary circumstance.²⁵⁴ In other words, an AI system developed to predict shoplifting and deployed by a private company may not fall within the scope of the prohibition.

ii. AI practices that are based solely on profiling of a natural person or on assessing their personality traits and characteristics

- 3.83** Article 5(1)(d) only prohibits AI practices that are solely based on profiling of a natural person or on assessing their personality traits and characteristics. Article 5(1)(d) does not apply to 'risk analytics that are not based on the profiling of individuals or on the personality traits and characteristics of individuals, such as AI systems using risk analytics to assess the likelihood of financial fraud by undertakings on the basis of suspicious transactions or various kinds of risk analytic tools'.²⁵⁵
- 3.84** The notion of **profiling** must be interpreted in accordance with the definition set out in Article 4(4) GDPR²⁵⁶, which defines profiling as 'any form of automated processing of personal data consisting of the use of personal data to evaluate certain personal aspects relating to a natural person, in particular to analyse or predict aspects concerning that natural person's performance at work, economic situation, health, personal preferences, interests, reliability, behaviour, location or movements'.²⁵⁷ The reference to the definition in the GDPR is appropriate since both frameworks aim to mitigate risks to fundamental rights arising from the use of such techniques. It will operate alongside Article 11(1) LED, pursuant to which Member States have to prohibit decisions based solely on automated processing, including profiling, which produces an adverse legal effect concerning the data subject or significantly affects him or her unless it is authorized by law and provides appropriate safeguards for the rights and freedoms of the data subject.
- 3.85** The Commission Guidelines underline that the concept of **group profiling**, which refers to the construction and the application of a descriptive profile for a given group must also be

²⁵⁰ Commission Guidelines on prohibited artificial intelligence practices (n 14) 71.

²⁵¹ Ibid.

²⁵² Article 2(3) AIA.

²⁵³ Commission Guidelines on prohibited artificial intelligence practices (n 14) 71.

²⁵⁴ Ibid.

²⁵⁵ Recital 42 AIA.

²⁵⁶ Article 3(52) AIA.

²⁵⁷ Article 4(4) GDPR.

considered as group profiles may later be applied to a specific individual.²⁵⁸ According to the Commission, whenever an AI system makes prediction and applies such a (group) profile to a specific individual, this constitutes profiling of the person and may, therefore, fall within the scope of the prohibition.²⁵⁹ This is justified by the fact that the prohibition aims to protect every individual in respect of whom the risk of committing a criminal offence is being predicted or assessed.²⁶⁰

The very vague notion of ‘**personality traits and characteristics**’ is intended to refer to elements, such as nationality, place of birth, place of residence, number of children, level of debt, or type of car.²⁶¹ Personality traits and characteristics hence constitute a broad category of characteristics related to a particular natural person, for which there is no generally agreed taxonomy.²⁶² This may include practices that also qualify as profiling, as defined by Article 4(4) GDPR, as well as those that fall below the thresholds of this legislative definition. Article 5(1)(d) leaves open the question whether practices that come within its scope need to be ‘solely’ based on assessment or personality traits and assessments.²⁶³ Recital 42 however implies that this criterion also applies to the assessment of personality traits or characteristics.²⁶⁴

The term ‘solely’ is critically important as Article 5(1)(d) applies only to risk assessments based exclusively on the two practices mentioned above. The prohibition does not apply to an AI system used to support the human assessment of the involvement of a person in a criminal activity, which is already based on objective and verifiable facts directly linked to a criminal activity.²⁶⁵ The notion of ‘solely automated processing’ in Article 22(1) GDPR and related interpretations may also be helpful in interpreting this notion.²⁶⁶ This criterion has been interpreted as requiring meaningful human oversight of a decision rather than just a token gesture.²⁶⁷

The judicial interpretation of this criterion will be key in determining the scope of Article 5(1)(d) overall. Indeed, the higher the threshold associated with the interpretation of ‘solely’, the narrower the scope of this prohibition.

iii. Exceptions

The last sentence of Article 5(1)(d) provides that the prohibition does not encompass AI systems **used to support the human assessment** of the involvement of a person in criminal activity that is already based on objective and verifiable facts directly linked to a criminal activity. According to the Commission Guidelines, the notion of ‘human assessment’ ought to

²⁵⁸ Commission Guidelines on prohibited artificial intelligence practices (n 14) 67. European Union Agency for Fundamental Rights, *Preventing Unlawful Profiling Today and in the Future: A Guide*. (Publications Office 2018) 21.

²⁵⁹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 68.

²⁶⁰ Ibid 67.

²⁶¹ Recital 42 AIA.

²⁶² Commission Guidelines on prohibited artificial intelligence practices (n 14) 68.

²⁶³ Indeed, the word ‘solely’ could be taken to only relate to ‘profiling’ or also to assessments based on ‘personality traits and characteristics’.

²⁶⁴ This interpretation has also been embraced in Commission Guidelines on prohibited artificial intelligence practices (n 14) 67.

²⁶⁵ Article 5(1)(d) AIA.

²⁶⁶ See further Case C-634/21 *SCHUFA* EU:C:2023:957.

²⁶⁷ WP29 Guidelines on Automated individual decision making and Profiling (n 213) 21.

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be interpreted in line with established judicial interpretations of ‘human intervention’ in EU law.²⁶⁸ In *Ligue des Droits Humains*, the CJEU established that human assessment must rely on objective criteria to evaluate an output of an automated system.²⁶⁹

- 3.90** The Commission Guidelines emphasize that there may also be **other situations** in which a risk assessment is not based solely on profiling or the evaluation of personality traits or characteristics; however, the consideration of other elements must be substantial and meaningful in order to justify the conclusion that the prohibition does not apply.²⁷⁰ In particular, the existence of certain pre-established objective and verifiable facts may justify that conclusion.²⁷¹ Recital 42 clarifies that risk analytics that are not based on the profiling of individuals or on the personality traits and characteristics of individuals—such as AI systems using risk analytics to assess the likelihood of financial fraud by undertakings based on suspicious transactions, or risk analytic tools to predict the likelihood of the localization of narcotics or illicit goods by customs authorities, for example, on the basis of known trafficking routes—fall outside the scope of the prohibition.²⁷²
- 3.91** Although these systems fall outside the scope of Article 5(1)(d), they still qualify as high-risk AI systems if they are intended to be used by or on behalf of law enforcement authorities or by EU institutions, bodies, offices, or agencies in support of law enforcement authorities.²⁷³
- 3.92** The prohibition in Article 5(1)(d) also does not apply to the **analysis of systemic crime patterns and geographic crime prediction**.²⁷⁴ The provision explicitly permits the use of AI systems in the investigation of crimes that have already taken place, restricting its prohibition solely to the use of such systems for the purpose of predicting potential future offences. However, distinguishing between AI systems designed for location-based risk analysis and those generating individualized predictions may prove challenging in practice—particularly in scenarios where location-derived risk scores are subsequently integrated into assessments of specific individuals. The Commission Guidelines recommend that such activity falls within the scope of the prohibition.²⁷⁵

VII. Article 5(1)(e): Facial Recognition

- 3.93** Article 5(1)(e) creates an absolute prohibition for the placing on the market, the putting into service for this specific purpose, or the use of AI systems that create or expand facial recognition databases through the untargeted scraping of facial images from the internet or closed-circuit television (‘CCTV’) footage.

²⁶⁸ Commission Guidelines on prohibited artificial intelligence practices (n 14) 70. See in particular Case C-817/19 *Ligue des droits humains* EU:C:2022:491.

²⁶⁹ See further Melanie Fink and Michèle Finck, ‘Reasoned A(I)Dministration: Explanation Requirements in EU Law and the Automation of Public Administration’ (2022) 47 *European Law Review* 376.

²⁷⁰ Commission Guidelines on prohibited artificial intelligence practices (n 14) 69.

²⁷¹ *Ibid.*

²⁷² Recital 42 AIA.

²⁷³ Article 6(2) and Point 6(a) of Annex III AIA.

²⁷⁴ Article 5(1)(d) and Recital 42 AIA.

²⁷⁵ Commission Guidelines on prohibited artificial intelligence practices (n 14) 73.

A. Overview

AI systems that create or expand facial recognition databases through the untargeted scraping of facial images from the internet or CCTV footage are deemed to add ‘to the feeling of mass surveillance and can lead to gross violations of fundamental rights, including the right to privacy.’²⁷⁶ There was, indeed, ample controversy around the usage of facial recognition techniques around the time that the regulation was negotiated.²⁷⁷ For example, a number of American cities had proceeded to ban usage of the technology in the late 2010s.²⁷⁸ The creation or expansion of facial recognition databases through the untargeted scraping of facial images from the internet or CCTV footage is, therefore, regarded as a significant risk to fundamental rights—even a gross violation—a description uniquely applied to this specific prohibition, highlighting that the co-legislators viewed it as an especially severe threat to fundamental rights and, by extension, to EU values.²⁷⁹ Notably, however, this prohibition was not part of the original draft regulation and was not proposed by the Council, but rather introduced by the Parliament mandate.²⁸⁰

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B. Analysis

Four cumulative conditions need to be met for an AI practice to be caught by Article 5(1)(e). First, the practice must constitute the ‘placing on the market’, ‘the putting into service for this specific purpose’, or the ‘use’ of an AI system. This criterion was examined in depth in relation to Article 5(1)(a), and the reader is referred to that analysis. It must, however, be noted that Article 5(1)(e) only governs AI systems that are put into service for the **specific purpose** of the creation or expansion of facial recognition databases through the untargeted scraping of facial images from the internet or CCTV footage.²⁸¹ This forms an important contrast to the prohibitions in Articles 5(1)(a)–(c), which apply to AI systems put into service generally. Neither the regulation nor the Commission Guidelines elaborate on the appropriate interpretation of this criterion. It can be reasonably assumed that what is required is the intent to pursue the specific objective of using an AI system to create or expand facial recognition databases through the untargeted scraping of facial images from the internet or CCTV footage. This intent may undoubtedly originate with the provider, who, as defined by the AI Act, is the natural or legal person putting the system into service.²⁸² However, it is worth wondering whether such intent could also arise from the deployer, especially given that ‘putting into service’ encompasses the initial supply of the AI system for direct

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²⁷⁶ Recital 43 AIA.

²⁷⁷ Evan Selinger and Woodrow Hartzog, ‘The Inconsistency of Facial Surveillance’ (2019) 66 *Loyola Law Review* 101; Peter Dauvergne, ‘Facial Recognition Technology for Policing and Surveillance in the Global South: A Call for Bans’ (2022) 43 *Third World Quarterly* 2325.

²⁷⁸ Kate Conger, Richard Fausset and Serge F Kovalski, ‘San Francisco Bans Facial Recognition Technology’ *The New York Times* (14 May 2019) <<https://www.nytimes.com/2019/05/14/us/facial-recognition-ban-san-francisco.html>> accessed 3 February 2025.

²⁷⁹ Article 2 TEU provides that ‘the Union is founded on the values of respect for human dignity, freedom, democracy, equality, the rule of law and respect for human rights, including the rights of persons belonging to minorities. These values are common to the Member States in a society in which pluralism, non-discrimination, tolerance, justice, solidarity and equality between women and men prevail.’

²⁸⁰ Four Columns AI Act (n 6) line 189b.

²⁸¹ Article 5(1)(d) AIA.

²⁸² Article 3(3) AIA.

use by the deployer.²⁸³ This raises important interpretive questions about the attribution of purpose within the provider-deployer relationship that must be assessed on a case-by-case basis.

- 3.96** Second, it must be used for the purpose of creating or expanding facial recognition databases. Third, the means to populate the database are through AI tools for untargeted scraping. Fourth, the sources of the images are either from the internet or CCTV footage.²⁸⁴ AI systems that fall short of meeting the requirements in Article 5(1)(e) may qualify as high-risk AI systems.²⁸⁵

i. The creation or expansion of facial recognition databases

- 3.97** Article 5(1)(e) targets systems used to either create or expand facial recognition databases. Facial recognition databases encompass any collection of data, or information²⁸⁶, that is specially organized for rapid search and retrieval by a computer that is capable of matching a human face from a digital image or video frame against a database of faces, comparing it to images in the database and determining whether there is a likely match between the two.²⁸⁷ Including the expansion of facial recognition databases was essential to ensure that databases established before the prohibition became binding on 2 February 2025 remain within the scope of the regulation, as their potential to infringe upon fundamental rights persists regardless of when they were created.²⁸⁸

ii. The untargeted scraping of facial images

- 3.98** Article 5(1)(e) only prohibits the creation or expansion of facial recognition databases using AI systems when it is done through the untargeted scraping of facial images. The Commission Guidelines draw an analogy between ‘untargeted scraping’ and a ‘vacuum cleaner, absorbing as much data and information as possible, without targeting specifically and individually intended subject(s) of the scraping.’²⁸⁹ Importantly, the respect of opt-out internet protocols, such as robot.txt, a text file that tells web robots (like search engine crawlers) which parts of a website they are allowed to access and which to avoid, does not affect the untargeted nature of the scraping.²⁹⁰ Reverse engineering image search engines are deemed to be targeted scraping.²⁹¹

- 3.99** Importantly, Article 5(1)(e) only covers facial image data and excludes other types of biometric material. Facial data, indeed, qualifies as biometric data, which is defined by Article 3(34) as ‘personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person, such as facial images or dactyloscopic data.’²⁹² This definition underlines that data protection law will also be applicable to such processing.

²⁸³ Article 3(11) AIA.

²⁸⁴ Article 5(1)(e) AIA and Commission Guidelines on prohibited artificial intelligence practices (n 14) 77.

²⁸⁵ See further Article 6(2) and Point 1(a) of Annex III AIA.

²⁸⁶ In contrast to data, information by definition has informational value.

²⁸⁷ Commission Guidelines on prohibited artificial intelligence practices (n 14) 78.

²⁸⁸ Article 113(3)(a) AIA.

²⁸⁹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 78.

²⁹⁰ Ibid. ‘Robots.Txt’, *Wikipedia* (2025) <<https://en.wikipedia.org/w/index.php?title=Robots.txt&oldid=1295385607>> accessed 27 June 2025.

²⁹¹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 79.

²⁹² Article 3(34) AIA.

iii. From the internet or CCTV footage

Third, the scraping must be carried out from **internet or CCTV footage**. While neither term is explicitly defined, they can be interpreted based on commonly accepted definitions. The Commission Guidelines provide examples of CCTV footage, such as images acquired by surveillance cameras operated in places like airports, streets, or parks.²⁹³

The prohibition does not apply to AI systems which harvest large amounts of facial images from the internet to build AI models that generate new images about fictitious persons because such systems are deemed to not result in the recognition of real persons.²⁹⁴ Whether such synthetic data really never qualifies as personal data is, however, open to debate.²⁹⁵ When such data appreciably resembles existing persons, it also falls within the scope of Article 50(4).²⁹⁶ The Commission Guidelines suggest that Article 5(1)(e) also does not apply to databases built prior to the entry into application of Chapter 2, namely, 2 February 2025, if these are not further expanded through AI-enabled untargeted scraping.²⁹⁷

VIII. Article 5(1)(f): Emotion Inference Systems in Employment and Education

Article 5(1)(f) formulates a prohibition regarding the placing on the market, the putting into service for this specific purpose, or the use of AI systems to infer emotions of a natural person in the areas of workplace and education institutions, except where the use of the AI system is intended to be put in place or into the market for medical or safety reasons.

A. Overview

Emotion recognition systems have faced widespread criticism for their unreliability, with many qualifying them as pseudoscience.²⁹⁸ Indeed, even the study accompanying the AI Act stated that to date, ‘inferring emotions, personality traits and other characteristics by means of biometric AI systems lacks scientific evidence. Any sentiment analysis software attempting to recognise human emotions is thus unproven. This is based on the fact that external expression does not always reflect inner emotional states accurately.’²⁹⁹ Recital 44

²⁹³ Commission Guidelines on prohibited artificial intelligence practices (n 14) 79.

²⁹⁴ Ibid.

²⁹⁵ Article 50 AIA. See further Michal S Gal and Orla Lynskey, ‘Synthetic Data: Legal Implications of the Data-Generation Revolution’ (2024) 109 Iowa Law Review 1087 <<https://ilr.law.uiowa.edu/volume-109-issue-3/2024/03/synthetic-data-legal-implications-data-generation-revolution>> accessed 22 March 2025.

²⁹⁶ See further Recital 134 AIA.

²⁹⁷ Commission Guidelines on prohibited artificial intelligence practices (n 14) 80.

²⁹⁸ Jay Stanley, ‘Experts Say “Emotion Recognition” Lacks Scientific Foundation’ (*American Civil Liberties Union*, 18 July 2019) <<https://www.aclu.org/news/privacy-technology/experts-say-emotion-recognition-lacks-scientific>> accessed 27 June 2025. For a general analysis, see Elisabeth Steindl, ‘Does the European Data Protection Framework Adequately Protect Our Emotions? Emotion Tech in Light of the Draft AI Act and Its Interplay with the GDPR’ (2022) 8 European Data Protection Law Review 311.

²⁹⁹ Commission, ‘Study to Support an Impact Assessment of Regulatory Requirements for Artificial Intelligence in Europe’ (Publications Office 2021) Final Report (D5), 39 <<https://data.europa.eu/doi/10.2759/523404>> accessed 4 February 2025.

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recognizes this as the basis for this prohibition, which raises the question as to why these systems are only prohibited in the contexts of the workplace and in education institutions.³⁰⁰ The co-legislators justified this sectoral approach by the imbalance of power that characterizes these settings, as well as the intrusive nature of these systems.³⁰¹

- 3.104** Article 5(1)(f) was not foreseen by the draft regulation and goes back to a suggestion by the European Parliament. In their joint opinion on the AI Act, the EDPB and EDPS had already underlined that the use of AI to infer the emotions of a natural person is highly undesirable and should be prohibited, except in certain well-specified use cases.³⁰²
- 3.105** The use of AI in the workplace is generally controversial because it raises significant concerns about surveillance, bias in decision-making, job displacement, and the erosion of workers' rights and autonomy.³⁰³ The Platform Workers Directive's ('PWD') rules on algorithmic management place limitations on the processing of personal data by means of automated monitoring systems or automated decision-making systems.³⁰⁴ While limited to the context of digital labour platforms, the prohibition has a much broader scope considering that it not only applies to AI systems but generally to automated monitoring systems or automated decision-making systems, which need not use AI. A European Parliament report recently recommended the adoption of another EU instrument on algorithmic management in the workplace.³⁰⁵
- 3.106** The integration of AI in education sparks heated debate due to concerns that it may undermine academic integrity, erode critical thinking and creativity, perpetuate algorithmic bias, infringe the privacy and data protection rights of students, and disrupt the teacher-student relationship.³⁰⁶
- 3.107** Emotion recognition systems that fall short of the prohibition in Article 5(1)(f) qualify as HRAIS.³⁰⁷ It is worth noting that deployers of emotion recognition systems are, by virtue of Article 50(3), additionally subject to an obligation to inform natural persons exposed thereto of the operation of the system.

³⁰⁰ Recital 44 AIA stating that there are 'serious concerns about the scientific basis of AI systems aiming to identify or infer emotions, particularly as expression of emotions vary considerably across cultures and situations, and even within a single individual. Among the key shortcomings of such systems are the limited reliability, the lack of specificity and the limited generalisability'. See also Commission Guidelines on prohibited artificial intelligence practices (n 14) 81.

³⁰¹ Recital 44 AIA.

³⁰² EDPB-EDPS Joint Opinion on the proposal for a Regulation of the AIA (n 194) para 35. According to the EDPB and the EDPS, these uses cases are health or research purposes.

³⁰³ Ifeoma Ajunwa, Kate Crawford and Jason Schultz, 'Limitless Worker Surveillance' (2017) 105 *California Law Review* 735; Jeremias Adams-Prassl and others, 'Regulating Algorithmic Management: A Blueprint' (2023) 14 *European Labour Law Journal* 124; Jeremias Adams-Prassl, 'Regulating Algorithms at Work: Lessons for a "European Approach to Artificial Intelligence"' (2022) 13 *European Labour Law Journal* 30.

³⁰⁴ Article 7(1)(a) PWD foresees that digital labour platforms shall, by means of automated monitoring systems or automated decision-making systems, process any personal data on the emotional or psychological state of a person performing platform work.

³⁰⁵ Parliament, 'Draft Report with recommendations to the Commission on digitalisation, artificial intelligence and algorithmic management in the workplace – shaping the future of work' (2025/2080(INL) <https://www.europarl.europa.eu/doceo/document/EMPL-PR-774283_EN.pdf> accessed 27 June 2025).

³⁰⁶ For an overview, see Margaret Bearman, Juliana Ryan and Rola Ajjawi, 'Discourses of Artificial Intelligence in Higher Education: A Critical Literature Review' (2023) 86 *Higher Education* 369 and Melissa Bond and others, 'A Meta Systematic Review of Artificial Intelligence in Higher Education: A Call for Increased Ethics, Collaboration, and Rigour' (2024) 21 *International Journal of Educational Technology in Higher Education* 4.

³⁰⁷ Article 6(2) and Point 1(c) of Annex III AIA. On the diverging terminology, see further below.

B. Analysis

Four cumulative conditions need to be met for the prohibition in Article 5(1)(f) to apply. First, the practice must constitute the ‘placing on the market’, ‘the putting into service for this specific purpose’, or the ‘use’ of an AI system. This criterion was examined in depth in relation to Article 5(1)(a), and the reader is referred to that analysis. It must, however, be noted that Article 5(1)(f) only governs AI systems that are put into service for the **specific purpose** of individual risk assessment and prediction of criminal offences. This forms an important contrast to the prohibitions in Articles 5(1)(a)–(c), which apply to AI systems put into service generally. Neither the regulation nor the Commission Guidelines elaborate on the appropriate interpretation of this additional criterion. It can be reasonably assumed that what is required is the intent to pursue the specific objective of using an AI system to infer emotions of a natural person in the areas of workplace and education institutions. This intent may undoubtedly originate with the provider, who, as defined by the AI Act, is the natural or legal person putting the system into service.³⁰⁸ However, one may equally question whether such intent could also arise from the deployer, especially given that ‘putting into service’ encompasses the initial supply of the AI system for direct use by the deployer.³⁰⁹ This raises important interpretive questions about the attribution of purpose within the provider-deployer relationship that must be assessed on a case-by-case basis.

Second, the system in question is an AI system to infer emotions. Third, the system is used in the workplace or education and training institutions. Fourth, the system is not intended to be put in place or into the market for medical or safety reasons.³¹⁰ Emotion recognition systems that fall short of meeting the criteria outlined by Article 5(1)(f) may qualify as high-risk AI systems.³¹¹ Article 50(3) furthermore requires that deployers of emotion recognition systems have to inform the natural persons exposed thereto of the operation of the system, unless the use of the system is permitted by law to detect, prevent, or investigate criminal offences, subject to appropriate safeguards for the rights and freedoms of third parties, and in accordance with EU law.

i. Inferring the emotions of a natural person

Article 5(1)(f) prohibits AI systems used to infer emotions of a natural person in the areas of workplace and education institutions. **Emotion inference systems** are not defined in the AI Act. This is remarkable as the regulation does, in turn provide a definition for emotion recognition systems.³¹² As per Article 3(39), an emotion recognition system (‘ERS’) is an AI system for the purpose of identifying or inferring emotions or intentions of natural persons based on their biometric data. This notion is relevant for the qualification of high-risk AI systems.³¹³ The use of differing terminology raises the question of whether the co-legislators intentionally employed distinct terms to encompass a broader range of systems. Specifically, this might include emotion recognition systems not reliant on the analysis of biometric data but also, for instance, those based on performance data. The fact that Article 3(39) defines

³⁰⁸ Article 3(3) AIA.

³⁰⁹ Article 3(11) AIA.

³¹⁰ Article 5(1)(f) AIA and Commission Guidelines on prohibited artificial intelligence practices (n 14) 81.

³¹¹ Article 6(2) and Point 1(c) of Annex III AIA.

³¹² The same diverging terminology can also be found in other language versions of the AIA. For instance, the French language version ‘reconnaissance’ and ‘inférer’, whereas the German language version speaks of ‘Emotionserkennungssystem’ and ‘ableiten’.

³¹³ See further Article 6(2) and Point 1(c) of Annex III AIA.

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an emotion recognition system as one that ‘infers’ emotion, as well as the fact that as per Recital 44, an emotion recognition system is an AI system ‘identifying or inferring emotions or intentions of natural persons’, indicate that these two terminologies should be understood as synonyms.³¹⁴ From a legislative quality standpoint, aligning the terminology would undoubtedly have been preferable to ensure clarity and consistency. Recital 18 lists examples of emotions that can be inferred from the emotion recognition system: ‘happiness, sadness, anger, surprise, disgust, embarrassment, excitement, shame, contempt, satisfaction and amusement.’³¹⁵ The Commission Guidelines recommend that the concept should be understood in a wide sense and not interpreted restrictively.³¹⁶

- 3.111** The **identification** of emotions occurs when the processing of the biometric data of a natural person allows a direct comparison and identification of an emotion with one that has been previously programmed into the emotion recognition system.³¹⁷ The **inference** of an emotion occurs by deducing information generated by analytical and other processes by the system itself, meaning that it is not solely based on collected data, but also inferred from other data.³¹⁸ Article 5(1)(f) only applies to emotion recognition and inference based on **biometric data**. Article 3(34) defines biometric data as personal data resulting from specific technical processing relating to the physical, physiological, or behavioural characteristics of a natural person, such as facial images or dactyloscopic data.³¹⁹

ii. Workplace and education institutions

- 3.112** Article 5(1)(f) only prohibits AI systems used to infer emotions of a natural person in the areas of workplace and education institutions. The usage of AI in these domains is, indeed, considered particularly risky considering existing power imbalances.³²⁰ Emotion inference systems used in other domains qualify as high-risk AI systems.³²¹ The Commission Guidelines underline that the concept of the **workplace** ought to be interpreted broadly to include ‘any setting where the work is performed and can vary widely based on the nature of the job, spanning from indoor office spaces, factories and warehouses to publicly accessible spaces like shops, stadiums or museums, to open-air sites or cars, as well as temporary or mobile work sites.’³²² The prohibition applies regardless of any legal classifications as an employee, contractor, trainee, volunteer, etc., and, for instance, also applies to candidates in a hiring process.³²³ Likewise, the notion of **education institutions** ought to be interpreted broadly. It includes both public and private institutions and there is no limitation regarding the types or ages of pupils or students, nor a specific environment.³²⁴

iii. Exceptions

- 3.113** Article 5(1)(f) includes an exception and does not apply to AI systems intended for deployment or market use for medical or safety purposes.³²⁵ Consequently, this leaves room

³¹⁴ See further Commission Guidelines on prohibited artificial intelligence practices (n 14) 82.

³¹⁵ Recital 18 AIA.

³¹⁶ Commission Guidelines on prohibited artificial intelligence practices (n 14) 82.

³¹⁷ Ibid.

³¹⁸ Ibid.

³¹⁹ Article 3(34) AIA. See also the discussion in Commission Guidelines on prohibited artificial intelligence practices (n 14) 82.

³²⁰ Recital 44 AIA.

³²¹ Article 6(2) and Point 1(c) of Annex III AIA.

³²² Commission Guidelines on prohibited artificial intelligence practices (n 14) 85.

³²³ Ibid.

³²⁴ Ibid.

³²⁵ See also Recital 44 AIA.

for emotion inference systems in workplaces or educational institutions, provided that their purpose is to identify safety risks or medical conditions. The interpretation of these purposes will play a critical role in determining the scope of the prohibition. According to the Commission Guidelines, this exception ought to be interpreted narrowly to ensure the regulation's objective of ensuring a high level of fundamental rights protection.³²⁶ Therapeutic uses should be understood to mean uses of CE-marked medical devices.³²⁷ Furthermore, the exception cannot include AI systems that detect general aspects of well-being, the general monitoring of stress levels, or AI systems intended to detect burnout or depression at the workplace or in education institutions.³²⁸ The notion of safety reasons can be understood as applying only in relation to the protection of life and health, and not to protect other interests, such as fraud or theft.³²⁹ Use cases that are caught by the exception need to be limited to what is strictly necessary and proportionate and accompanied by sufficient safeguards, such as motivated expert opinions.³³⁰

Importantly, Member States can keep or adopt laws, regulations, or administrative provisions that are more favourable to workers in terms of protecting their rights.³³¹ This is one of the few areas where Member States retain policy autonomy in relation to AI in the aftermath of the AIA.

IX. Article 5(1)(g): Biometric Categorization Based on Certain Characteristics

Article 5(1)(g) prohibits the placing on the market, the putting into service for this specific purpose, or the use of biometric categorization systems that categorize individually natural persons based on their biometric data to deduce or infer their race, political opinions, trade union membership, religious or philosophical beliefs, sex life or sexual orientation. The prohibition does not cover any labelling or filtering of lawfully acquired biometric datasets, such as images, based on biometric data or categorizing of biometric data in law enforcement.

A. Overview

Article 5(1)(g) prohibits biometric categorization systems, which categorize individual natural persons based on their biometric data to deduce a closed list of characteristics. Unlike the preceding prohibitions, Article 5(1)(g) lacks an explicitly stated rationale.³³² However, it appears evident that it is similarly rooted in the protection of fundamental rights, such as the right to human dignity as enshrined in Article 1 CFR. The Commission Guidelines, moreover, refer to the risks of unfair and discriminatory treatment that can result from such practices, such as when a service is denied to someone based on their race.³³³ Biometric

³²⁶ Commission Guidelines on prohibited artificial intelligence practices (n 14) 87.

³²⁷ *Ibid.*

³²⁸ *Ibid.*

³²⁹ *Ibid.*

³³⁰ *Ibid.*

³³¹ Article 2(11) AIA.

³³² Recital 30 AIA addresses biometric categorization systems but merely states that they ought to be prohibited.

³³³ Commission Guidelines on prohibited artificial intelligence practices (n 14) 90.

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classification can involve sorting individuals into groups based on physical traits like facial features, skin colour, or body shape. Some of these traits, such as race, are considered particularly sensitive or protected under EU anti-discrimination laws.³³⁴ Additionally, biometric classification can also be based on DNA or behavioural patterns, like typing rhythms or the way a person walks.³³⁵

- 3.117** Regarding **legislative history**, Article 5(1)(g) was not foreseen in the Commission draft. The current text goes back to a proposal from the European Parliament, which suggested a prohibition of biometric categorization systems ‘that categorise natural persons according to sensitive or protected attributes or characteristics or based on the inference of those attributes or characteristics’ unless the AI system is used for approved therapeutic uses on the basis of informed consent.³³⁶ The reference to ‘sensitive or protected attributes’ would have been noticeably broader but also have afforded less legal certainty compared to the current formulation. Considering that consent is widely viewed as an inadequate means to legitimize the processing of (sensitive) personal data it is unsurprising that this condition was omitted.³³⁷

B. Analysis

- 3.118** Five cumulative conditions need to be met for an AI practice to be caught by Article 5(1)(g). First, the practice must constitute the ‘placing on the market’, ‘the putting into service for this specific purpose’, or ‘the use’ of an AI system. This criterion was examined in depth in relation to Article 5(1)(a), and the reader is referred to that analysis. It must, however, be noted that Article 5(1)(g) only governs AI systems that are put into service for the **specific purpose** of categorizing individually natural persons based on their biometric data to deduce or infer their race, political opinions, trade union membership, religious or philosophical beliefs, sex life, or sexual orientation. This forms an important contrast to the prohibitions in Articles 5(1)(a)–(c), which apply to AI systems put into service generally. Neither the regulation nor the Commission Guidelines elaborate on the appropriate interpretation of this additional criterion. It can be reasonably assumed that what is required is the intent to pursue the specific objective of using an AI system to categorize individually natural persons based on their biometric data to deduce or infer their race, political opinions, trade union membership, religious or philosophical beliefs, sex life or sexual orientation. This intent may undoubtedly originate with the provider, who, as defined by the AI Act, is the natural or legal person putting the system into service.³³⁸ However, one may equally question whether such intent could also arise from the deployer, especially given that ‘putting into service’ encompasses the initial supply of the AI system for direct use by the deployer.³³⁹ This raises important interpretive questions about the attribution

³³⁴ See further Council Directive 2000/43/EC of 29 June 2000 implementing the principle of equal treatment between persons irrespective of racial or ethnic origin [2000] OJ L180/22.

³³⁵ Commission Guidelines on prohibited artificial intelligence practices (n 14) 92. See further Article 29 Data Protection Working Party, ‘Opinion 3/2012 on developments in biometric technologies’ (WP 193, 27 April 2012) 16–17. The Group refers here to ‘soft recognition’, i.e. ‘detection of behaviour or specific needs of people’.

³³⁶ Four Columns AI Act (n 6) line 182a.

³³⁷ Elettra Bietti, ‘Consent as a Free Pass: Platform Power and the Limits of the Informational Turn’ (2020) 40 Pace Law Review 310.

³³⁸ Article 3(3) AIA.

³³⁹ Article 3(11) AIA.

of purpose within the provider-deployer relationship that must be assessed on a case-by-case basis.

Second, the system must qualify as a biometric categorization system under Article 3(40).³⁴⁰ Third, individual persons must be categorized individually based on their biometric data. Fourth, the categorization based on biometric data must lead to a deduction or inference of their race, political opinions, trade union membership, religious or philosophical beliefs, sex life, or sexual orientation.³⁴¹ If a biometric identification system does not meet these criteria, it can be placed on the market, put into service for this specific purpose, or used; however, it will be classified as a high-risk AI system, and must therefore comply with the obligations set out in Chapter III.³⁴² Article 50(3) furthermore requires that deployers of biometric categorization systems have to inform the natural persons exposed thereto of the operation of the system, unless the use of the system is permitted by law to detect, prevent, or investigate criminal offences, subject to appropriate safeguards for the rights and freedoms of third parties, and in accordance with EU law.

i. Biometric categorization systems

A biometric categorization system is ‘an AI system for the purpose of assigning natural persons to specific categories on the basis of their biometric data, unless it is ancillary to another commercial service and strictly necessary for objective technical reasons.’³⁴³

Biometric data is ‘personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person, such as facial images or dactyloscopic data.’³⁴⁴ This notion has to be interpreted in light of the definition of biometric data in Article 4(14) GDPR.³⁴⁵ In EU data protection law, biometric data forms a subcategory of special category personal data, that is to say that it is considered particularly sensitive in nature and therefore benefits from additional related protections compared to ordinary personal data.³⁴⁶ There is, however, an important distinction between the two legislative provisions as Article 4(14) GDPR foresees that biometric data ‘allow[s] or confirm[s] the unique identification of that natural person’. Whereas this formulation was initially also included in the draft regulation, it was later dropped due to criticisms that this requirement makes the definition too narrow and fails to keep pace with technological developments.³⁴⁷

³⁴⁰ Article 3(40) AIA defines the biometric categorization system as follows: ‘an AI system for the purpose of assigning natural persons to specific categories on the basis of their biometric data, unless it is ancillary to another commercial service and strictly necessary for objective technical reasons’.

³⁴¹ Article 5(1)(g) AIA and Commission Guidelines on prohibited artificial intelligence practices (n 14) 91. The Communication breaks this down into five distinct criteria, distinguishing individual categorization and based on their personal data. However, the Commission’s subsequent analysis of Article 5(1)(g) AIA merges these two criteria into one.

³⁴² See further Article 6(2) and Point 1(b) of Annex III AIA.

³⁴³ Article 3(40) AIA.

³⁴⁴ Article 3(34) AIA.

³⁴⁵ Article 4(14) GDPR reads as follows: ‘Biometric data means personal data resulting from specific technical processing relating to the physical, physiological or behavioural characteristics of a natural person, which allow or confirm the unique identification of that natural person, such as facial images or dactyloscopic data’. The same definition can also be found in Article 3(13) LED and in Article 3(18) EUDPR.

³⁴⁶ See also Article 3(37) AIA.

³⁴⁷ See further Four Columns AI Act (n 6) line 160. Christiane Wendehorst, *The Proposal for an Artificial Intelligence Act COM(2021) 206 from a Consumer Policy Perspective: Part I: The Qualification of AI Systems and Practices as Prohibited or High-Risk* (Bundesministerium für Soziales, Gesundheit, Pflege und Konsumentenschutz 2021) 94.

ii. Individual categorization based on a closed list of characteristics

3.121 A biometric categorization system is a system that individually assigns natural persons to specific categories based on biometric data.³⁴⁸ The Commission Guidelines indicate that the individual categorization can be the ‘purpose or outcome’ of biometric data analysis.³⁴⁹ In any event, categorizations of entire groups are not subject to this prohibition.³⁵⁰ The individual biometric categorization needs to occur in order to deduce or infer race, political opinions, trade union membership, religious or philosophical beliefs, sex life, or sexual orientation. Neither the legislative text nor the Commission Guidelines explain the decision to use a closed list of criteria.

iii. Exceptions

3.122 Biometric categorization systems do not include those systems that are both **ancillary to another commercial service and strictly necessary for objective technical reasons**.³⁵¹ This means that the feature cannot, for objective technical reasons, be used without the principal service, and the integration of that feature or functionality is not a means to circumvent the regulation.³⁵² Examples of biometric categorization falling outside the scope of Article 5(1) (g) are filters used on online marketplaces that categorize facial or body features if they can be used only in relation to the principal service, namely selling a product by allowing the consumer to preview the display of the product to make a purchase decision.³⁵³ Another example is that of filters used on online social network services, which categorize facial or body features to allow users to add or modify pictures or videos.³⁵⁴

3.123 Importantly, Recital 30 furthermore provides that the prohibition ‘should not cover the lawful labelling, filtering or categorisation of biometric data sets acquired in line with Union or national law according to biometric data, such as the sorting of images according to hair colour or eye colour, which can *for example* be used in the area of **law enforcement**’.³⁵⁵ This leaves open the question whether the use of such data is exclusively permitted in the domain of law enforcement, or whether law enforcement is solely one of the multiple possible domains where such data can be used.³⁵⁶ The divergence in text likely stems from differing positions during the trilogue negotiations. However, from a legislative quality perspective, this inconsistency is regrettable and indicative of the repeated issues the text presents from this perspective.

3.124 According to the Commission Guidelines, this exemption is justified by the fact that labelling on the basis of protected sensitive information can be ‘necessary for high-quality data, precisely to prevent discrimination’, for instance, to comply with the obligations imposed by Articles 10 and 17 for high-risk AI systems.³⁵⁷ Yet, the exclusion of law enforcement from the

³⁴⁸ Recital 14 AIA.

³⁴⁹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 93. (‘If this is not the purpose or outcome of the biometric categorisation, the prohibition does not apply, for example if a whole group is categorised without looking at the individual!’)

³⁵⁰ Ibid.

³⁵¹ These two conditions need to be fulfilled cumulatively. See also Commission Guidelines on prohibited artificial intelligence practices (n 14) 92.

³⁵² Ibid.

³⁵³ Ibid.

³⁵⁴ Ibid.

³⁵⁵ Recital 30 AIA (my own emphasis).

³⁵⁶ The Commission Guidelines implicitly seems to suggest that this exemption is only available in the area of law enforcement. See further Commission Guidelines on prohibited artificial intelligence practices (n 14) 95.

³⁵⁷ Ibid 94. See further Articles 10 and 17 AIA.

prohibition also creates a gap where application of such systems with particularly high relevance to fundamental rights and rule of law principles are likely to occur.³⁵⁸

Recitals 40 and 41, moreover, create exemptions for Ireland and Denmark, respectively from the application of Article 5(1)(g) regarding biometric categorization systems for activities in the field of police cooperation and judicial cooperation in criminal matters, which is based on Protocol 21 of the TEU and TFEU that governs Ireland's and Denmark's limited participation in the area of freedom, security, and justice.³⁵⁹

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X. Article 5(1)(h): Real-Time Remote Biometric Identification Systems in Publicly Accessible Spaces for the Purposes of Law Enforcement

Article 5(1)(h) prohibits the use of real-time remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement, unless and in so far as such use is strictly necessary for one of the following objectives: (i) the targeted search for specific victims of abduction, trafficking in human beings or sexual exploitation of human beings, as well as the search for missing persons; (ii) the prevention of a specific, substantial and imminent threat to the life or physical safety of natural persons or a genuine and present or genuine and foreseeable threat of a terrorist attack; (iii) the localization or identification of a person suspected of having committed a criminal offence, for the purpose of conducting a criminal investigation or prosecution or executing a criminal penalty for offences referred to in Annex II and punishable in the Member State concerned by a custodial sentence or a detention order for a maximum period of at least four years.

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A. Overview

In addition to the prohibitions outlined above, the AI Act also creates detailed and complex rules regarding the use of biometric identification systems, such as facial recognition systems, which have been the subject of much controversy.³⁶⁰ The regulation of these use cases of artificial intelligence was one of the thorniest issues during the legislative procedure, including the trilogues.³⁶¹ The result is a qualified prohibition of RTRBIS in publicly accessible spaces for the purposes of law enforcement. The qualified prohibition in Article 5(1)(h) must be interpreted in conjunction with the rules outlined in Articles 5(2)–(8). These rules are complemented by various definitions as well as no less than six recitals.³⁶² This structure

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³⁵⁸ Marie Eneman and others, 'The Sensitive Nature of Facial Recognition: Tensions between the Swedish Police and Regulatory Authorities I' (2022) 27 Information Polity 219; Maša Galič and Lonneke Stevens, 'Regulating Police Use of Facial Recognition Technology in the Netherlands: The Complex Interplay between Criminal Procedural Law and Data Protection Law' (2023) 14 New Journal of European Criminal Law 459.

³⁵⁹ See further Ester Herlin-Karnell, 'Denmark and the European Area of Freedom, Security and Justice: A Scandinavian Arrangement' (2013) 5 Amsterdam Law Forum 95.

³⁶⁰ Denise Almeida, Konstantin Shmarko and Elizabeth Lomas, 'The Ethics of Facial Recognition Technologies, Surveillance, and Accountability in an Age of Artificial Intelligence: A Comparative Analysis of US, EU, and UK Regulatory Frameworks' (2022) 2 AI and Ethics 377; Selinger and Hartzog (n 277).

³⁶¹ Luca Bertuzzi, 'Leading MEPs Make Counter-Proposal on AI Rulebook's Law Enforcement Chapter' (*Euractiv*, 21 November 2023) <<https://www.euractiv.com/section/artificial-intelligence/news/leading-meps-make-counter-proposal-on-ai-rulebooks-law-enforcement-chapter/>> accessed 3 February 2025.

³⁶² Recitals 32, 33, 38, 39, 40 and 41 AIA.

is **confusing**, as Article 5(1)(h) is followed by seven subsequent paragraphs that elaborate on the conditions and criteria for a closed list of exceptions to this prohibition. Furthermore, perhaps the most important aspect of this regime is only made explicit in Article 5(5), namely that a **Member State law** needs to authorize these systems for these rules to apply.³⁶³

3.128 Article 5(1)(h) assumes that RTRBIS, such as facial recognition technology, in public spaces for the purposes of law enforcement pose **risks to fundamental rights**, and that these risks are greater than those associated with the biometric identification systems listed in Point 1(a) of Annex III, which qualify as HRAIS.³⁶⁴ If the use of a RTRBIS is permissible under the complex regime established in Chapter II, they also are HRAIS.³⁶⁵ In addition, specific rules regulate the retrospective, non-real-time use of remote biometric identification systems for law enforcement. Article 26(10) formulates specific safeguards that deployers must respect when using such post-remote biometric identification systems.³⁶⁶

3.129 The interplay between Articles 5(1)(h)–(8) and the **high-risk AI systems** regime requires clarification for subsequent analysis.³⁶⁷ Article 5(1)(h) establishes a qualified prohibition on real-time remote biometric identification systems deployed for law enforcement purposes in publicly accessible spaces. While such systems cannot be used by deployers within the EU, providers may nonetheless place them on the market or put them into service, as Article 5(1)(h) exclusively governs use rather than the placing on the market and putting into service of these systems.³⁶⁸ Moreover, systems falling within the exceptions listed in Article 5(1)(h)(i)–(iii) remain permissible for deployer use. In these two scenarios, these systems constitute high-risk AI systems pursuant to the combined reading of Article 6(2) and Point 1(a) of Annex III, which designates any ‘remote biometric identification system’ as high risk, including those within the scope of Article 5(1)(h).³⁶⁹ Indeed, all remote biometric identification systems qualify as high risk in view of the risks that they pose.³⁷⁰ The Chapter III obligations for HRAIS are predominantly provider-centric, ensuring compliance with requisite standards, even in cases where deployers may lawfully utilize real-time remote biometric identification systems for law enforcement in publicly accessible spaces.³⁷¹ Thus, while there is a material overlap between the systems that fall within the scope of the non-prohibited practices under Article 5(1)(h) and those caught by Point 1(a) of Annex I, there is little overlap *ratione personae* regarding these applicable norms as most rules in Chapter III apply to providers.³⁷²

³⁶³ See also Recital 37 AIA (‘such use in the territory of a Member State in accordance with this Regulation should only be possible where and in as far as the Member State concerned has decided to expressly provide for the possibility to authorise such use in its detailed rules of national law. Consequently, Member States remain free under this Regulation not to provide for such a possibility at all or to only provide for such a possibility in respect of some of the objectives capable of justifying authorised use identified in this Regulation.’)

³⁶⁴ These include remote biometric identification systems; AI systems intended to be used for biometric categorization, and emotion recognition systems.

³⁶⁵ Point 1(a) of Annex III AIA. This provision, however, foresees one exception to this qualification, namely that AI systems intended to be used for biometric verification the sole purpose of which is to confirm that a specific natural person is who they claim to be.

³⁶⁶ Article 26(10) AIA.

³⁶⁷ See further Chapter III AIA.

³⁶⁸ See further Articles 2(1)(a)–(c) and Articles 3(3) and (4) AIA.

³⁶⁹ Note that Point 1(a) of Annex III AIA foresees an exception to this classification rule, namely that ‘AI systems intended to be used for biometric verification the sole purpose of which is to confirm that a specific natural person is the person he or she claims to be’ are not high-risk AI systems.

³⁷⁰ Recital 54 AIA.

³⁷¹ It should be noted that specific obligations also arise for providers of biometric identification systems, namely the logging obligations outlined in Article 12(3) AIA.

³⁷² There are, however, some exceptions to this, such as Article 26(1) AIA, which is examined further below.

This two-tiered approach to RTRBIS implies an underlying assumption that **real-time** identification systems pose greater risks to fundamental rights due to their temporal dimension. This assumption can, however, be questioned, as the Orwellian risk of persistent tracking is equally pronounced in real-time and non-real-time systems.³⁷³ These systems can all have a deterring effect on the exercise of fundamental rights, such as the right to freedom of assembly. Similarly, the right to privacy and the sense of being under surveillance are significantly affected irrespective of whether data is analyzed in real-time or the following day.³⁷⁴ The joint opinion of the EDPB and the EDPS on the draft regulation, indeed, stressed that the distinction between ‘real-time’ and non-real-time biometric identification systems is futile from the perspective of fundamental rights.³⁷⁵ NGOs also unsuccessfully drew attention to this point throughout the legislative procedure.³⁷⁶

Yet, according to the regulation, RTRBIS are deemed particularly intrusive to the rights and freedoms of the concerned persons, to the extent that it may affect the private life of a large part of the population, evoke a feeling of constant surveillance, and indirectly dissuade the exercise of the freedom of assembly and other fundamental rights.³⁷⁷ The technical inaccuracies of these systems can, moreover, lead to biased results and entail discriminatory effects.³⁷⁸ Such bias can occur in particular with respect to age, ethnicity, race, sex or disabilities.³⁷⁹ The regulation, moreover, highlights that the ‘immediacy of the impact and the limited opportunities for further checks or corrections in relation to the use of such systems operating in real-time carry heightened risks for the rights and freedoms of the persons concerned in the context of, or impacted by, law enforcement activities.’³⁸⁰ As a result, the AI Act only permits the use of real-time biometric identification systems in publicly accessible spaces for the purposes of law enforcement ‘where the use is strictly necessary to achieve a substantial public interest, the importance of which outweighs the risks.’³⁸¹

Biometric identification methods were **one of the most controversial topics** during the legislative procedure. The European Commission’s Impact Assessment identified biometric surveillance as a fundamental rights concern, laying the groundwork for this prohibition. It stressed that artificial intelligence can give rise to unprecedented opportunities for

³⁷³ European Parliament Directorate General for Parliamentary Research Services, *Person Identification, Human Rights and Ethical Principles: Rethinking Biometrics in the Era of Artificial Intelligence*. (Publications Office 2021) <<https://data.europa.eu/doi/10.2861/384495>> accessed 7 February 2025.

³⁷⁴ Nathalie A Smuha and Karen Yeung, ‘The European Union’s AI Act: Beyond Motherhood and Apple Pie?’ in Nathalie A Smuha (ed), *The Cambridge Handbook of the Law, Ethics and Policy of Artificial Intelligence* (1st edn, CUP 2025).

³⁷⁵ EDPB-EDPS Joint Opinion on the proposal for a Regulation of the AIA (n 194) 2.

³⁷⁶ See e.g. European Digital Rights (EDRI) and others, ‘Prohibit all Remote Biometric Identification (RBI) in publicly accessible spaces’ <<https://edri.org/wp-content/uploads/2022/05/Prohibit-RBI-in-publicly-accessible-spaces-Civil-Society-Amendments-AI-Act-FINAL.pdf>> accessed 4 February 2025.

³⁷⁷ Recital 32 AIA; Michael O’Flaherty, ‘Opinions · Facial Recognition Technology and Fundamental Rights’ (2020) 6 European Data Protection Law Review 170; Daragh Murray, ‘Facial Recognition and the End of Human Rights as We Know Them?’ (2024) 42 Netherlands Quarterly of Human Rights 145; European Union Agency for Fundamental Rights, ‘Facial recognition technology: fundamental rights considerations in the context of law enforcement’ (FRA, 21 November 2019) <https://fra.europa.eu/sites/default/files/fra_uploads/fra-2019-facial-recognition-technology-focus-paper.pdf> accessed 4 February 2025.

³⁷⁸ Recital 32 AIA. See further David Leslie, ‘Understanding Bias in Facial Recognition Technologies’ (Zenodo 2020) <<https://zenodo.org/record/4050457>> accessed 27 June 2025; Gideon Christian, ‘The New Jim Crow: Unmasking Racial Bias in AI Facial Recognition Technology within the Canadian Immigration System’ (Social Science Research Network, 29 May 2024) <<https://papers.ssrn.com/abstract=4846967>> accessed 4 February 2025.

³⁷⁹ Recital 32 AIA.

³⁸⁰ Ibid.

³⁸¹ Recital 33 AIA.

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‘indiscriminate or mass surveillance, profiling and scoring of citizens and significant intrusion into people’s privacy and other fundamental rights’, such as through the use of remote biometric identification systems in publicly accessible spaces.³⁸² This was deemed to threaten not only data protection and privacy, but also freedom of expression, association, and assembly, which in turn might have a chilling effect on democracy.³⁸³ A moratorium on facial recognition technology was foreseen in a leaked version of the AI White Paper.³⁸⁴

3.133 The Commission draft would have prohibited the use of real-time remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement, unless and in so far as such use is strictly necessary for objectives including (i) the targeted search for potential victims of crime, including missing children; (ii) preventing a specific or imminent threat to the life or physical safety of natural persons or a terrorist attack, or (iii) searching for potential criminals in the context of the European Arrest Warrant.³⁸⁵ The other two co-legislators suggested extensive revisions to this text. The Parliament mandate would have banned any use of real-time remote biometric identification systems in publicly accessible spaces irrespective of the purpose they are used for and have dropped any exceptions to this ban.³⁸⁶ The Council would have only banned the use of such systems in publicly accessible spaces ‘by law enforcement authorities or on their behalf’.³⁸⁷ It would have kept the exceptions with some changes, such as the removal of the explicit reference to missing children in objective (i), an extension of objective (ii) through the inclusion of threats to critical infrastructure, life, health, and physical safety of persons, as well as the deletion of the requirement that the threat needed to be ‘imminent’, a rewriting and partial extension of objective (iii).³⁸⁸

3.134 Biometric surveillance remained the most controversial topic during the trilogues.³⁸⁹ It was reported that political agreement on this specific point was only reached orally during the trilogues, with no written text agreed upon by all parties. The oral agreement stipulated that real-time surveillance would, in principle, be prohibited, while remote systems used by law enforcement would be classified as high risk.³⁹⁰ Reportedly, following the trilogues, the Spanish Council presidency presented a written text to Members of the European Parliament that diverged from the oral agreement, being significantly more permissive of biometric surveillance.³⁹¹ As a result, this point needed to be renegotiated after the trilogues.³⁹² This legislative history helps situate the complex nature of the rules governing biometric identification systems, particularly the inclusion of numerous recitals—a common technique in EU

³⁸² Commission Impact Assessment (n 9) 18.

³⁸³ Ibid.

³⁸⁴ Natasha Foote, ‘LEAK: Commission Banks on Biogas to Protect EU Farmers from Energy Crisis’ (*Euractiv*, 22 February 2022) <<https://www.euractiv.com/section/agriculture-food/news/leak-commission-banks-on-bio-gas-to-protect-eu-farmers-from-energy-crisis/>> accessed 4 February 2025.

³⁸⁵ Article 5(1)(d) dAIA.

³⁸⁶ Four Columns AI Act (n 6) line 186.

³⁸⁷ Ibid line 825.

³⁸⁸ Ibid.

³⁸⁹ Steffen Stierle, ‘AI Act: Kommt Big Brother durch die Hintertür?’ (*Tagesspiegel Background Digitalisierung & KI*, 11 January 2024) <<https://background.tagesspiegel.de/digitalisierung-und-ki/briefing/ai-act-kommt-big-brother-durch-die-hintertuer>> accessed 4 February 2025.

³⁹⁰ Ibid.

³⁹¹ Ibid.

³⁹² Ibid.

law-making employed when consensus on adopting a specific rule into the legislative text cannot be reached.³⁹³

Article 5(1)(h) applies as *lex specialis* to Article 10 LED, which sets out conditions for the processing of special categories of personal data in the context of law enforcement.³⁹⁴ Alice Giannini and Sarah Tas have argued that this has significant consequences as many Member States have merely transposed Article 10 into national law without adopting specific legislation for law enforcement authorities processing of sensitive biometric data.³⁹⁵ Yet, according to the EDPB, simple transposition cannot serve as a legal basis for biometric data processing due to insufficient precision and foreseeability, requiring specific national provisions that many jurisdictions currently lack.³⁹⁶ A potential regulatory conflict emerges as a result of this as RTRBIS might be permissible under AI Act while remaining prohibited or unregulated under national law implementing the LED.³⁹⁷ Consequently, the AI Act may effectively circumvent existing restrictions arising from inadequate LED implementation at the national level.³⁹⁸ This could be remedied through the national enabling legislation Member States have to adopt by virtue of Article 5(5).

B. Analysis

Three cumulative conditions need to be fulfilled for the prohibition in Article 5(1)(h) to apply. First, there must be a use of a real-time remote biometric identification system. Second, it must be used in in publicly accessible spaces, and third, for law enforcement purposes.³⁹⁹

i. The use of a 'real-time' remote biometric identification system

A **remote biometric identification system** is an AI system for the purpose of identifying natural persons without their active involvement, typically at a distance through the comparison of a person's biometric data with the biometric data contained in a reference database.⁴⁰⁰ Facial recognition technology is probably the best-known example of a biometric identification system.⁴⁰¹ The regulation hence embraces a **functional definition**, which does

³⁹³ See further Michael Veale and Lilian Edwards, 'Clarity, Surprises, and Further Questions in the Article 29 Working Party Draft Guidance on Automated Decision-Making and Profiling' (2018) 34 Computer Law & Security Review 398.

³⁹⁴ Recital 38 AIA. Article 10 LED reads as follows: 'Processing of personal data revealing racial or ethnic origin, political opinions, religious or philosophical beliefs, or trade union membership, and the processing of genetic data, biometric data for the purpose of uniquely identifying a natural person, data concerning health or data concerning a natural person's sex life or sexual orientation shall be allowed only where strictly necessary, subject to appropriate safeguards for the rights and freedoms of the data subject, and only: (a) where authorised by Union or Member State law; (b) to protect the vital interests of the data subject or of another natural person; or (c) where such processing relates to data which are manifestly made public by the data subject'.

³⁹⁵ Alice Giannini and Sarah Tas, 'AI Act and the Prohibition of Real-Time Biometric Identification' (*Verfassungsblog*, 10 December 2024) <<https://verfassungsblog.de/ai-act-and-the-prohibition-of-real-time-biometric-identification/>> accessed 27 June 2025.

³⁹⁶ EDPB, 'Guidelines 05/2022 on the use of facial recognition technology in the area of law enforcement' (version 2.0, 26 April 2023) <https://www.edpb.europa.eu/our-work-tools/our-documents/guidelines/guidelines-052022-use-facial-recognition-technology-area_en> accessed 27 June 2025.

³⁹⁷ Giannini and Tas (n 395).

³⁹⁸ Ibid.

³⁹⁹ Article 5(1)(h) AIA.

⁴⁰⁰ See also Article 3(41) AIA.

⁴⁰¹ For an overview, see further Neil Selwyn and others, 'Facial Recognition Technology: Key Issues and Emerging Concerns' in Rita Matulionyte and Monika Zalnieriute (eds), *The Cambridge Handbook of Facial Recognition in the Modern State* (1st edn, CUP 2024).

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not focus on the system's technical features but rather its purpose, namely the identification of natural persons without their active involvement, 'irrespective of the particular technology, processes or types of biometric data used'.⁴⁰² RBIS are typically used to perceive multiple persons or their behaviour simultaneously in order to facilitate the identification of natural persons without their active involvement.⁴⁰³

- 3.138** **Biometric identification** means the automated recognition of physical, physiological, behavioural, or psychological human features for the purpose of establishing the identity of a natural person by comparing biometric data of that individual to biometric data of individuals stored in a database.⁴⁰⁴ RBIS hence do not include biometric verification systems that merely serve to confirm the identity of one particular person, such as for authentication purposes.⁴⁰⁵ Biometrics are a tool used 'to recognise or verify the identity of a person based on their external appearance or behavioural characteristics'.⁴⁰⁶
- 3.139** Article 5(1)(h) only applies to 'real-time' **remote** biometric identification systems. Remoteness implies the ability of biometric systems to identify individuals without their active involvement, typically at a distance through the comparison of a person's biometric data with the biometric data contained in a reference database.⁴⁰⁷ Biometric systems to confirm the identity of a natural person for the sole purpose of having access to a service, unlocking a device, or having security access to premises cannot be remote,⁴⁰⁸ as they are deemed to only have a minor impact on fundamental rights.⁴⁰⁹
- 3.140** Article 5(1)(h) also only covers '**real-time**' biometric identification systems, to the exclusion of biometric identification systems that are not real time. The use of quotation marks around the term *real-time* is unusual in legislative drafting, as it may suggest that the term is not to be understood in its literal sense, thereby introducing unnecessary ambiguity. A real-time remote biometric identification system is a 'remote biometric identification system, whereby the capturing of biometric data, the comparison and the identification all occur without a significant delay, comprising not only instant identification, but also limited short delays in order to avoid circumvention'.⁴¹⁰ This ought to be distinguished from post-remote biometric identification systems, which refer to a 'remote biometric identification system other than a real-time remote biometric identification system'.⁴¹¹ Deployers of post-remote biometric identification systems must meet the deployer obligations outlined in Article 26(10), *inter alia*, to request authorizations, document usage, and report on the use of these systems.
- 3.141** Crucially, however, the legislative text provides no definition of '**without a significant delay**'. According to the Commission Guidelines, this notion needs to be interpreted on a case-by-case basis and in general, 'a delay is significant at least when the person is likely to have left the place where the biometric data was taken'.⁴¹² The Guidelines also specify that when a law

⁴⁰² Recital 17 AIA.

⁴⁰³ *Ibid.*

⁴⁰⁴ Article 3(35) and Recital 38 AIA.

⁴⁰⁵ Pursuant to Article 3(41) AIA, identification happens where a natural person is identified without their active involvement and typically (though not necessarily) at a distance through the comparison of a person's biometric data with the biometric data contained in a reference database.

⁴⁰⁶ Commission Study to Support AI Impact Assessment (n 299) 39.

⁴⁰⁷ Article 3(41) AIA.

⁴⁰⁸ Recital 15 AIA.

⁴⁰⁹ Recital 17 AIA.

⁴¹⁰ Article 3(42) AIA.

⁴¹¹ Article 3(43) AIA.

⁴¹² Commission Guidelines on prohibited artificial intelligence practices (n 14) 101.

enforcement authority covertly takes a picture of a person via a mobile device and submits it to a database for immediate search, depending on the circumstances, this may fall under the prohibition.⁴¹³ The textual grounding for limiting this to pictures taken covertly is unclear as covertness does not necessarily correspond to remoteness.

It is important to clarify the **relationship with other AI systems subject to the AI Act**. Biometric identification systems in particular need to be distinguished from **biometric verification systems**, which enable the automated one-to-one verification, including authentication, of the identity of a natural person by comparing their biometric data to previously provided biometric data.⁴¹⁴ Biometric identification systems are also different from **biometric categorization systems**, which are AI systems used for the purpose of assigning natural persons to specific categories on the basis of their biometric data, unless it is ancillary to another commercial service and strictly necessary for objective technical reasons.⁴¹⁵ **Emotion recognition systems**, which use biometric data to identify or infer emotions or intentions of natural persons are also not biometric identification systems.⁴¹⁶

Remarkably, Article 5(1)(h) only prohibits the **use** of real-time biometric identification systems, meaning that the prohibition only concerns deployers.⁴¹⁷ This forms an important contrast with all other provisions in Article 5, which apply to providers as well as deployers.⁴¹⁸

ii. Publicly accessible spaces

Article 5(1)(h) prohibits ‘real-time’ remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement. A ‘publicly accessible space’ is a ‘publicly or privately owned physical place accessible to an undetermined number of natural persons, regardless of whether certain conditions for access may apply, and regardless of the potential capacity restrictions.’⁴¹⁹ The determination of whether a space is publicly accessible always needs to be done on a case-by-case basis.⁴²⁰

A publicly accessible space can be used for any activity, such as commerce, transport, entertainment, sport, or leisure.⁴²¹ A space should not be considered to be publicly accessible where ‘access is limited to specific and defined natural persons through either Union or national law directly related to public safety or security or through the clear manifestation of will by the person having the relevant authority over the space.’⁴²² Company and factory premises, as well as offices and workplaces that are intended to be accessed only by relevant employees and service providers are not publicly accessible spaces.⁴²³ Recital 19 explicitly qualifies ‘prisons and border control’ as spaces that are not publicly accessible.⁴²⁴ The exclusion of ‘border control’ may be surprising as within the European Union, where most

⁴¹³ Ibid.

⁴¹⁴ Article 3(36), Article 6(2) and Point 1(a) of Annex III AIA.

⁴¹⁵ Article 3(40), Article 5(1)(g), Article 6(2), Article 50(3) and Point 1(a) of Annex III AIA.

⁴¹⁶ Article 3(39), Article 5(1)(g) and Article 50(3) AIA.

⁴¹⁷ Deployers of AIS are defined by Article 3(4) AIA as ‘a natural or legal person, public authority, agency or other body using an AI system under its authority except where the AI system is used in the course of a personal non-professional activity’. The concept of use was defined in the analysis related to Article 5(1)(a) AIA.

⁴¹⁸ See further Articles 5(1)(a)–(g) AIA.

⁴¹⁹ Article 3(44) AIA.

⁴²⁰ Commission Guidelines on prohibited artificial intelligence practices (n 14) 103.

⁴²¹ Recital 19 AIA.

⁴²² Ibid.

⁴²³ Ibid.

⁴²⁴ Ibid.

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Member States participate in the Schengen Agreement⁴²⁵, most border areas are in fact publicly accessible. The notion of ‘border control’ hence ought to be interpreted restrictively to not include a broader area at a border but only the premises where controls are carried out, in line with the formulations in other linguistic versions of the regulation.⁴²⁶ In practice this may be difficult to delineate with precision, such as where a highway runs along a border and police occasionally carries out controls on the sidelines. According to the Commission Guidelines, a border crossing point is not a publicly accessible space, while the street leading to the border crossing point or a forest in the vicinity normally is.⁴²⁷

- 3.146** Recital 19 furthermore usefully clarifies that the factual possibility of access alone, ‘such as an unlocked door or an open gate in a fence’, does not imply that the space is publicly accessible when there are indications or circumstances suggesting the contrary, like signs prohibiting or restricting access.⁴²⁸ The regulation acknowledges that some areas ‘may comprise both publicly accessible and non-publicly accessible spaces, such as the hallway of a private residential building necessary to access a doctor’s office or an airport.’⁴²⁹ Importantly, online spaces are not covered, ‘as they are not physical spaces.’⁴³⁰

iii. For the purposes of law enforcement

- 3.147** Article 5(1)(h) only prohibits the use of ‘real-time’ remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement. Prior to the publication of the draft regulation, the Council of Europe Commissioner for Human Rights had cautioned that ‘Member States should apply the highest level of scrutiny when using AI systems in the context of law enforcement, especially when engaging in methods such as predictive or preventive policing.’⁴³¹ Given the potential consequences for individuals, law enforcement is a particularly sensitive domain.

- 3.148** Law enforcement is defined as ‘activities carried out by law enforcement authorities or on their behalf for the prevention, investigation, detection or prosecution of criminal offences or the execution of criminal penalties, including safeguarding against and preventing threats to public security.’⁴³² The definition is aligned with Article 2(2)(d) GDPR, meaning that the CJEU may build on its case law under this provision for future interpretations of Article 4(46).⁴³³ Law enforcement authorities are public authorities competent for the prevention, investigation, detection, or prosecution of criminal offences, or the execution of criminal penalties, including the safeguarding against and the prevention of threats to public security, as well as any other body or entity entrusted by Member State law to exercise public authority and public powers for the purposes of the prevention, investigation, detection or

⁴²⁵ The Schengen *acquis* Agreement between the Governments of the States of the Benelux Economic Union, the Federal Republic of Germany and the French Republic on the gradual abolition of checks at their common borders [2000] OJ L239/13.

⁴²⁶ The French text refers to ‘le contrôle aux frontières’, whereas the German version speaks of ‘Grenzkontrollbereiche’.

⁴²⁷ Commission Guidelines on prohibited artificial intelligence practices (n 14) 102.

⁴²⁸ Recital 19 AIA.

⁴²⁹ Ibid.

⁴³⁰ Ibid.

⁴³¹ Council of Europe Commissioner for Human Rights, ‘Unboxing Artificial Intelligence: 10 Steps to Protect Human Rights’ (*Council of Europe Bookshop*, 2019b) 11 <<https://book.coe.int/en/commissioner-for-human-rights/9664-unboxing-artificial-intelligence-10-steps-to-protect-human-rights.html>> accessed 6 February 2025.

⁴³² Article 3(46) AIA.

⁴³³ See, e.g. Case C-175/20 *SIA ‘SS’ v Valsts ieņēmumu dienests* EU:C:2022:124 and Case C-439/19 *B v Latvijas Republikas Saeima* EU:C:2021:504.

prosecution of criminal offences or the execution of criminal penalties, including the safeguarding against and the prevention of threats to public security.⁴³⁴ The AI Act relies on the same definition of the law enforcement authority as the LED.⁴³⁵ Yet, it remains unsettled which actors are precisely included in this broad definition.⁴³⁶ In any event, tax authorities and safety directorates do not qualify as law enforcement authorities in EU law.⁴³⁷ In practice, the distinction between law enforcement and **national security**, which benefits from the exemption in Article 2(3), will sometimes be challenging to delineate, such as in relation to terrorist activity or large-scale organized crime.⁴³⁸

The limitation of the prohibition to law enforcement alone has been subject to criticism, as using this technology, whether by law enforcement or other actors, almost inevitably infringes on fundamental rights due to the disproportionate processing of large amounts of sensitive data from many people to identify a few individuals.⁴³⁹ AI systems used outside of enforcement purposes—such as commercial applications recommending hairstyles, eye-wear, or clothing in virtual fitting rooms—fall outside the scope of the prohibition.

iv. Grounds based on which Member States can authorize the use of RTRBIS

Article 5(1)(h) foresees **three exceptions** to its qualified prohibition. Articles 5(2)–(7) lay down the conditions and safeguards applicable to these exceptions. Importantly, Article 5(1)(h)(i)–(iii) do not in themselves constitute a legal basis for the use of RTRBIS by law enforcement in publicly accessible spaces. Rather, only a **Member State law** that fulfils the requirements in Article 5(2)–(7) can allow such use.⁴⁴⁰ Consequently, in the absence of Member State legislation, such use has been prohibited since 2 February 2025.⁴⁴¹

The objectives for which the use of RTRBIS for law enforcement purposes in publicly accessible spaces is allowed must be strictly, exhaustively, and narrowly defined, and appear when there is a ‘strict necessity’ to achieve ‘a substantial public interest’, which ‘outweighs the risks’ posed to fundamental rights.⁴⁴² Member States can only allow such use in three ‘exhaustively listed and narrowly defined situations, where the use is strictly necessary to achieve a substantial public interest, the importance of which outweighs the risks.’⁴⁴³ This wording indicates that courts are expected to interpret the discretion granted to Member States narrowly. Nonetheless, this poses a difficult balancing task, as the expressly enumerated exemptions are, by their nature, broadly framed.

First, national legislation can provide for the use of ‘real-time’ remote biometric identification systems in publicly accessible spaces for law enforcement purposes, specifically for the **targeted search for specific victims of abduction, trafficking in human beings or sexual exploitation of human beings, as well as the search for missing persons**.⁴⁴⁴ A critical issue is whether this exception applies exclusively to criminal conduct or extends more broadly to

⁴³⁴ Article 3(45) AIA.

⁴³⁵ Article 3(45) AIA and Article 4(7)(a) LED. See also Article 2(2)(d) GDPR.

⁴³⁶ Luca Tosoni and Lee A Bygrave, ‘Definitions’ in Eleni Kosta and Franziska Boehm (eds), *The EU Law Enforcement Directive (LED)* (1st edn, OUP 2024).

⁴³⁷ Case C-175/20 *Valsts ienēmumu dienests* and Case C-439/19 *Latvijas Republikas Saeima*.

⁴³⁸ See further Article 2(3) AIA.

⁴³⁹ Smuha and others (n 128) 25.

⁴⁴⁰ Article 5(2) AIA and Commission Guidelines on prohibited artificial intelligence practices (n 14) 105.

⁴⁴¹ Article 113 AIA.

⁴⁴² Recital 33 AIA.

⁴⁴³ Ibid.

⁴⁴⁴ Article 5(1)(h)(i) AIA.

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other activities. Article 5(1)(h)(i) refers to the ‘targeted search or victims [...] as well as the search for missing persons’. In contrast, Recital 33 refers to ‘the search for certain victims of crime including missing persons’.⁴⁴⁵ The Commission Guidelines interpret the reference to missing persons as not necessitating a link to criminal activity so to include missing persons who may have voluntarily left home or those who are unaccounted for due to medical conditions, such as dementia.⁴⁴⁶

3.153 Second, real-time biometric identification systems for the purposes of law enforcement in publicly accessible spaces can be used for the **prevention of a specific, substantial and imminent threat to the life or physical safety of natural persons or a genuine and present or genuine and foreseeable threat of a terrorist attack**.⁴⁴⁷

3.154 An **imminent threat to life or the physical safety of natural persons** can result from a serious disruption of critical infrastructure⁴⁴⁸, where the disruption or destruction of such critical infrastructure would result in an imminent threat to life or the physical safety of a person, including through serious harm to the provision of basic supplies to the population or to the exercise of the core function of the State. According to the Commission Guidelines, the definition and assessment of what constitutes an imminent threat to life or physical safety are determined by each Member State according to its national laws.⁴⁴⁹ The Guidelines, moreover, clarify that threats need not be linked to criminal activity considering that they can be non-intentional. What matters is that the threat is clearly defined, individualized, and concrete, in that it should not be hypothetical or relate to certain dangers in general.⁴⁵⁰

3.155 This formulation is notably broad, particularly with terms like ‘terrorism’, which has long been used as an undefined legal concept.⁴⁵¹ In addition, one may wonder whether **terrorism** is included at all, considering that the CJEU has established that national security, an area that altogether falls outside the scope of the AI Act by virtue of Article 2(3), ‘encompasses the prevention and punishment of activities capable of seriously destabilising the fundamental constitutional, political, economic or social structures of a country and, in particular, of directly threatening society, the population or the State itself, such as terrorist activities’.⁴⁵² There is an important distinction between the phrases ‘genuine and present’ and ‘genuine and foreseeable’ terrorist attack, which can significantly impact the interpretation and application of the provisions. The Commission Guidelines offer a rare insight into the legislative history of the regulation in stating that these formulations were inspired by the CJEU’s case law on data retention and passenger name record measures aimed at safeguarding national security against terrorist attacks.⁴⁵³ In *SpaceNet* (2022), the CJEU held that in those contexts, ‘a threat to national security must be genuine and present, or at the very least, foreseeable,

⁴⁴⁵ Recital 33 AIA.

⁴⁴⁶ Commission Guidelines on prohibited artificial intelligence practices (n 14) 106.

⁴⁴⁷ Article 5(1)(h)(ii) AIA.

⁴⁴⁸ As defined in Article 2(4) of the Directive (EU) 2022/2557 of the European Parliament and of the Council of 14 December 2022 on the resilience of critical entities and repealing Council Directive 2008/114/EC [2022] OJ L333/163.

⁴⁴⁹ Commission Guidelines on prohibited artificial intelligence practices (n 14) 107.

⁴⁵⁰ Ibid 108.

⁴⁵¹ Ben Saul, *Defining Terrorism in International Law* (Reprinted, OUP 2007); Reuven Young, ‘Defining Terrorism: The Evolution of Terrorism as a Legal Concept in International Law and Its Influence on Definitions in Domestic Legislation’ (2006) 29 Boston College International and Comparative Law Review 23; Clive Walker, *Terrorism and the Law* (OUP 2011).

⁴⁵² Joined Cases C-511/18, C-512/18 and C-520/18 *La Quadrature du Net and Others* EU:C:2020:791, para 135

⁴⁵³ Commission Guidelines on prohibited artificial intelligence practices (n 14) 108.

which presupposes that sufficiently concrete circumstances have arisen.⁴⁵⁴ Again, the assessment regarding the existence and seriousness of such a threat ought to be made by the Member State concerned.⁴⁵⁵ Considering that this exemption available to Member States is concerned with prevention, it cannot be used to locate or identify a particular person. However, it can be used to locate 'several persons linked to the same threat, if there are concrete indications that those persons plan to commit a terrorist attack, but it is not clear where'.⁴⁵⁶ The Commission's emphasis on the plural raises the question of whether it can also be used to locate a single individual, which would make sense if there is a suspicion that a single person was involved.

Third, real-time biometric identification systems for the purposes of law enforcement in publicly accessible spaces can be used for the **localization or identification of a person suspected of having committed a criminal offence, for the purpose of conducting a criminal investigation or prosecution or executing a criminal penalty** for offences referred to in Annex II and punishable in the Member State concerned by a custodial sentence or a detention order for a maximum period of at least four years.⁴⁵⁷ Pursuant to Recital 33, 'such a threshold for the custodial sentence or detention order in accordance with national law contributes to ensuring that the offence should be serious enough to potentially justify the use of 'real-time' remote biometric identification systems'.⁴⁵⁸

Annex II includes a closed list of the following offenses: terrorism, human trafficking, sexual exploitation of children and child pornography, illicit trafficking in narcotic drugs or psychotropic substances, weapons, munitions, or explosives, murder, grievous bodily injury, illicit trade in human organs or tissue, trafficking in nuclear or radioactive materials, kidnapping, illegal restraint, hostage-taking, crimes under the jurisdiction of the International Criminal Court, unlawful seizure of aircraft or ships, rape, environmental crimes, organized or armed robbery, sabotage, and participation in a criminal organization involved in any of the aforementioned offenses. The list of criminal offences provided in Annex II is based on the 32 criminal offences listed in the Council Framework Decision 2002/584/JHA.⁴⁵⁹ This decision addresses criminal offences, such as rape, for which no harmonized definition exists among Member States, resulting in significant jurisdictional divergences within the European Union.⁴⁶⁰ Notably, certain Member States require the absence of consent as a constitutive element of rape, while others, including France, do not.⁴⁶¹ Consequently, regulatory fragmentation persists between Member States, creating disparate legal standards and potentially undermining the uniform application of criminal justice across the Union.⁴⁶²

⁴⁵⁴ Joined Cases C-793/19 and C-794/19 *SpaceNet* EU:C:2022:702, para 93.

⁴⁵⁵ Commission Guidelines on prohibited artificial intelligence practices (n 14) 109.

⁴⁵⁶ *Ibid.*

⁴⁵⁷ This is without prejudice to Article 9 GDPR for the processing of biometric data for purposes other than law enforcement.

⁴⁵⁸ Recital 33 AIA.

⁴⁵⁹ *Ibid.* It does not, however, list all of these offences included here, 'taking into account that some of those offences are, in practice, likely to be more relevant than others, in that the recourse to "real-time" remote biometric identification could, foreseeably, be necessary and proportionate to highly varying degrees for the practical pursuit of the localisation or identification of a perpetrator or suspect of the different criminal offences listed and having regard to the likely differences in the seriousness, probability and scale of the harm or possible negative consequences'.

⁴⁶⁰ *Giannini and Tas* (n 395).

⁴⁶¹ *Ibid.*

⁴⁶² *Ibid.*

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- 3.158** Hidden away in Article 5(3) is the rule that **no decision that produces an adverse legal effect** on a person may be taken based solely on the output of the ‘real-time’ remote biometric identification system.⁴⁶³ The rules in Article 5(1)(h)(i)–(iii) need to be interpreted accordingly.

v. Interaction with data protection law

- 3.159** Article 5(1)(h) will often interact with **data protection law**. Biometric identification systems require the processing of biometric data.⁴⁶⁴ Considering that biometric data by definition qualifies as personal data⁴⁶⁵, data protection law—particularly the LED—governs its use in the context of law enforcement. Article 5(1)(h) is *lex specialis* in respect of the rules on the processing of biometric data in Article 10 LED, thus regulating such use and the processing of biometric data involved in an exhaustive manner.⁴⁶⁶ This implies that such use and processing should be possible only insofar as it is compatible with the AI Act, leaving no scope outside that framework for the competent authorities, where they act for purpose of law enforcement, to use such systems and process such data in connection thereto on the grounds listed in Article 10 LED.⁴⁶⁷ Importantly, the AI Act does not create a new legal basis for the processing of personal data under Article 8 LED.⁴⁶⁸ Any processing of biometric data and other personal data involved in the use of AI systems for biometric identification, other than in connection to the use of real-time remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement needs to continue to comply with all requirements of Article 10 LED.⁴⁶⁹ Denmark and Ireland are not bound by these rules as far as they pertain to police cooperation and judicial cooperation in criminal matters in line with Protocols No 21 and 22 to the European Treaties.⁴⁷⁰

XI. Article 5(2): Rules Regarding the Use of Real-Time Biometric Identification Systems in Publicly Accessible Spaces for the Purposes of Law Enforcement

- 3.160** Article 5(2) outlines the rules governing the application of the three grounds on which Member States can permit the use of RTRBIS.⁴⁷¹ It provides that the use of real-time remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement for any of the objectives referred to in Article 5(1)(h) can be deployed for the purposes set out in that point only to confirm the identity of the specifically targeted individual. It also has to take into account the following elements: (a) the nature of the situation giving rise to the possible use, in particular the seriousness, probability and scale of the harm that would be caused if the system were not used; and (b) the consequences of the use of the system for the rights and freedoms of all persons concerned, in particular the seriousness, probability and scale of those consequences.

⁴⁶³ Article 5(3) AIA.

⁴⁶⁴ See further Article 3(34) AIA.

⁴⁶⁵ See Article 4(1) GDPR and Article 3(1) LED.

⁴⁶⁶ Recital 38 AIA.

⁴⁶⁷ Ibid.

⁴⁶⁸ Ibid.

⁴⁶⁹ Recital 39 AIA.

⁴⁷⁰ Article 6a of Protocol No 21(n 4). See also Recital 40 and 41 AIA.

⁴⁷¹ Article 5(1)(h) and 5(5) AIA.

Such uses furthermore must comply with necessary and proportionate safeguards and conditions in relation to the use in accordance with the national law authorizing the use thereof, in particular as regards the temporal, geographic and personal limitations. In particular, the use of such systems can only be authorized if the law enforcement authority has completed a fundamental rights impact assessment and has registered the system in the EU database.⁴⁷² However, in duly justified cases of urgency, the use of such systems may be commenced without the registration in the EU database, provided that such registration is completed without undue delay.

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A. Overview

Article 5(1)(h) creates a qualified prohibition for the use of real-time biometric identification systems in publicly accessible spaces for the purposes of law enforcement. Articles 5(2)–(7) formulate safeguards and conditions that apply where Member States use RTRBIS for one of the objectives listed in Article 5(1)(h)(i)–(iii). Article 5(2) establishes a floor rather than a ceiling of protection. It authorizes Member States to adopt more restrictive regulations governing remote biometric identification systems, provided such measures comply with Union law.⁴⁷³

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In terms of its **legislative history**, Article 5(2) was already included in the draft regulation. However, the restriction to the confirmation of the identity of specifically targeted individuals was introduced only during the trilogue negotiations.⁴⁷⁴ In the trilogues, further safeguards were also introduced, such as the need for a fundamental rights impact assessment, as well as the registration of the system in the EU database.⁴⁷⁵

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B. Analysis

i. Confirmation of the identity of a specifically targeted individual

Article 5(2) provides that biometric identification systems in publicly accessible spaces for the purposes of law enforcement can only be ‘**deployed**’ in the areas listed in Article 5(1)(h)(i)–(iii) to confirm the identity of the specifically targeted individual. The notion of deployment is not defined and is only used in this provision as well as a few recitals.⁴⁷⁶ Considering that the deployer is the natural or legal person using an AI system, it can be assumed that deployment is intended as a synonym for ‘use’.⁴⁷⁷

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The reference to ‘**specifically targeted individual**’ in the singular suggests that only one person can be targeted at a time even when an investigation is carried out conjointly for two or multiple persons that may have fallen victim to the same crime. This serves ‘as an additional safeguard for the fundamental rights limiting the risk of indiscriminate surveillance’.⁴⁷⁸ This rule implies that RTRBIS can only be used to search for specific individuals

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⁴⁷² See further Articles 27 and 49 AIA.

⁴⁷³ Article 5(5) AIA.

⁴⁷⁴ Four Columns AI Act (n 6) line 190.

⁴⁷⁵ Ibid line 193.

⁴⁷⁶ Recitals 3, 34, 131 AIA.

⁴⁷⁷ On the notion of the deployer, see further Article 3(4) AIA. This interpretation is also supported by the meaning of the verb ‘to deploy’ in the recitals making use of it.

⁴⁷⁸ Commission Guidelines on prohibited artificial intelligence practices (n 14) 113.

for which the law enforcement authorities have reasons to believe or are informed that they are involved in the scenarios listed in Article 5(1)(h)(i)–(iii).⁴⁷⁹ This requirement does not, however, imply that law enforcement authorities ‘need to know the identity of the individuals they are searching for before using the system.’⁴⁸⁰ The deployment of a RTRBIS can only be authorized for ‘targeted individuals’, in order to balance the seriousness of the situation and the harm resulting from not using the system against the impact the use has on fundamental rights, in particular to avoid mass surveillance.⁴⁸¹ If law enforcement authorities have ‘factual indications and information’ about a planned terrorist attack by a terrorist group at a specific time and place but do not know who will execute the plan, a RTRBIS can be used to identify the offender from the terrorist group, provided that the law enforcement authorities have constituted a reference database containing the biometric data of the individuals forming part of the terrorist group.⁴⁸² While this example relates only to Article 5(1)(h)(ii), the same rationale should arguably be extended to Article 5(1)(h)(i) and Article 5(1)(h)(iii).⁴⁸³ The Guidelines furthermore specify that the notion ‘confirming the identity’ may also include the localization of the person in question.⁴⁸⁴

ii. Elements to be considered

- 3.166** When determining the appropriateness of deploying remote biometric identification systems, law enforcement authorities must consider **the nature of the situation giving rise to the possible use**, in particular the seriousness, probability, and scale of the harm that would be caused if the system were not used.
- 3.167** Furthermore, the **consequences of the use of the system for the rights and freedoms of all persons concerned**, in particular the seriousness, probability and scale of those consequences must be taken into account in decisions whether to use remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement.
- 3.168** The Commission Guidelines provide that the **seriousness** criterion ought to be interpreted as referring to ‘a variation in degrees of interference with the fundamental rights at stake, which is linked to the principle of proportionality’, as well as that some interferences with fundamental rights are deemed more serious than others (without specifying which ones).⁴⁸⁵ The **scale** criterion in turn refers to the number but also categories of persons affected (such as children or vulnerable groups).⁴⁸⁶ **Probability** refers to the likelihood that an event will occur.⁴⁸⁷ Seriousness, scale, and probability all need to be assessed in the context of the fundamental rights impact assessment.⁴⁸⁸
- 3.169** The real-time use of RBIs needs to be clearly limited in terms of geographic scope, duration, and the targeted person.⁴⁸⁹ The choice of **geographic location** needs to be based on objective and non-discriminatory factors, which ‘implies that the geographic restriction applies to a clearly delineated boundary for which there are indications that the event will take

⁴⁷⁹ Ibid.

⁴⁸⁰ Ibid.

⁴⁸¹ Ibid.

⁴⁸² Ibid.

⁴⁸³ See further Article 5(1)(h)(i)–(iii) AIA.

⁴⁸⁴ Commission Guidelines on prohibited artificial intelligence practices (n 14) 113.

⁴⁸⁵ Ibid 114.

⁴⁸⁶ Ibid.

⁴⁸⁷ Ibid.

⁴⁸⁸ Ibid.

⁴⁸⁹ Article 5(2)(b) AIA.

place' and normally be smaller than an entire city.⁴⁹⁰ Regarding **personal scope**, the provision excludes 'the untargeted, indiscriminate identification of persons, without further indications of an incident'.⁴⁹¹ Regarding the **time limit**, the use needs to be limited to what is strictly necessary and an initial time-limited authorization may be extended in accordance with rules devised by the Member States. The period may be neither indefinite nor vague and needs to be determined based on the concrete indications that lead to the use of the system.⁴⁹²

iii. Compliance with necessary and proportionate safeguards and conditions

Furthermore, the use of remote biometric identification systems for the purposes of law enforcement must, moreover, foresee **necessary and proportionate safeguards and conditions** regarding the use in accordance with the national law authorizing the use thereof, in particular as regards the temporal, geographic and personal limitations. The national law authorizing the uses of RTRBIS in accordance with Article 5(5) must foresee such safeguards. Article 5(2), moreover, requires that a fundamental rights impact assessment has been carried out by the law enforcement authority.⁴⁹³ FRIAs are governed by Article 27, which sets out the requirements applicable to fundamental rights impact assessments for high-risk AI systems. As seen above, AI systems whose use is lawful subject to the criteria set out in Article 5(1)(h)–(8) qualify as high-risk AI systems and must abide with all related requirements. Article 27(1) creates a general obligation for deployers—either entities governed by public law or private entities providing public services—to perform an impact assessment on fundamental rights that may be affected by the use of such system.⁴⁹⁴ Related requirements are outlined in Article 27 and the reader is referred to the analysis of this provision for further detail. Article 27(5) obliges the AI Office to develop a template for a questionnaire, which will, in due course, shed further light on what the FRIA entails. Whereas Chapter II became applicable on 2 February 2025, Chapter III, including Article 27, will only become binding on 2 August 2026.⁴⁹⁵ The Commission Guidelines deem that, notwithstanding, the requirements for FRIA set out in Article 27 'should be implemented by the deployers of real-time RBI systems meeting the conditions to benefit from one or more of the exceptions in Article 5(1)(h)'.⁴⁹⁶ These Guidelines also set out in further detail what a FRIA should entail, while noting that this interpretation only applies to Article 5(2) and not high-risk AI systems generally, and only until the AI Office template is finalized.⁴⁹⁷ Considering that RTRBIS process personal data, the data controllers that operate them will also be required to carry out data protection impact assessments.⁴⁹⁸

Article 5(2) furthermore obliges deployers of RTRBIS to register their system in the EU database governed by Articles 49 and 71. In accordance with Article 49(3), deployers that are public authorities, Union institutions, bodies, offices or agencies (or persons acting on their behalf) have to register themselves, select the system, and register its use in the EU

⁴⁹⁰ Commission Guidelines on prohibited artificial intelligence practices (n 14) 114.

⁴⁹¹ Ibid.

⁴⁹² Ibid.

⁴⁹³ For a definition of law enforcement authorities, see further Article 3(45) AIA.

⁴⁹⁴ Excluded from this obligation are high-risk AI systems used in critical infrastructure. See further Article 27(1) AIA.

⁴⁹⁵ Article 113(2) and 113(3)(a) AIA.

⁴⁹⁶ Commission Guidelines on prohibited artificial intelligence practices (n 14) 115.

⁴⁹⁷ Ibid 115–118.

⁴⁹⁸ Article 27 LED, Article 35 GDPR or Article 39 EUDPR.

database for high-risk AI systems listed in Annex III. Whereas Chapter II became applicable on 2 February 2025, Chapter III, including Article 27, will only become binding on 2 August 2026.⁴⁹⁹ At the time of writing, this database, which the Commission must set up and maintain in collaboration with Member States, did not yet exist.⁵⁰⁰

- 3.172** In **duly justified cases of urgency**, the use of the RTRBIS may be commenced without the registration in the EU database, provided that such registration is completed without undue delay. The last sentence of Article 5(2) ties into the authorization system outlined in Article 5(3), as well as the urgency procedure governed by the same provision, both of which set out specific rules for addressing such scenarios. This authorization needs to be issued by a judicial authority or an independent administrative authority whose decision is binding. It is issued in the Member State in which the use is to take place. In duly justified situations of urgency, such authorization can be requested up until 24 hours after the use of the AI system. Member States may also issue blanket authorizations.⁵⁰¹

XII. Article 5(3): Prior Authorization Granted by a Judicial Authority or an Independent Administrative Authority

- 3.173** Articles 5(3) outlines the authorization procedures Member States need to respect if they decide to enable the use of real-time remote biometric identification systems in publicly accessible spaces for the purposes of law enforcement in the scenarios listed in Articles 5(1)(h)(i)–(iii).⁵⁰² The objective of Article 5(3) is to make sure that each use is necessary and proportionate to achieve any of the objectives listed in Articles 5(1)(h)(i)–(iii), and limited to what is strictly necessary concerning the temporal, geographic, and personal scope.⁵⁰³

A. Overview

- 3.174** Article 5(3) must be interpreted considering the broad discretion afforded to Member States concerning real-time remote biometric identification systems. Such systems may only be deployed under the conditions set out in Article 5(1)(h) where their use is expressly authorized by domestic legislation.⁵⁰⁴ Moreover, this national legislation must establish detailed procedural safeguards, including authorization requirements applicable to each individual instance of use, in accordance with the framework of national law. It has been pointed out that this deference to Member States could undermine the regulation's objective of protecting fundamental rights, particularly in those with lower levels of judicial independence, where it might be used to approve the use of real-time remote biometric identification systems without sufficient scrutiny.⁵⁰⁵

⁴⁹⁹ Article 113(2) and 113(3)(a) AIA.

⁵⁰⁰ Article 71(1) AIA.

⁵⁰¹ Article 5(5) AIA.

⁵⁰² See further Article 5(5) and Recital 37 AIA.

⁵⁰³ Commission Guidelines on prohibited artificial intelligence practices (n 14) 120.

⁵⁰⁴ See further Article 5(5) and Recital 37 AIA.

⁵⁰⁵ Hacker (n 19).

B. Analysis

Article 5(3) provides that each use of a ‘real-time’ remote biometric identification system in publicly accessible spaces for the purposes of law enforcement must be subject to a **prior authorization granted by a judicial authority or an independent administrative authority** whose decision is binding of the Member State in which the use is to take place, issued upon a reasoned request and in accordance with the detailed rules of national law referred to in Article 5(5). The Commission Guidelines recall the importance of **judicial independence** in relation to judicial authorities.⁵⁰⁶ They also refer to the CJEU case law on the interpretation of the notion of independence.⁵⁰⁷

Each use of an RBI system needs to be authorized.⁵⁰⁸ Furthermore, each use can only be authorized if it is **necessary and proportionate** to achieve one of the objectives outlined in Articles 5(1)(h)(i)–(iii). In EU law, an interference with fundamental rights and freedoms must always respect the essence of fundamental rights and freedoms.⁵⁰⁹

Article 5(3) also sets out the conditions under which related requests have to be examined. It provides that the **competent judicial authority or an independent administrative authority** whose decision is binding can grant the authorization only where it is satisfied, on the basis of objective evidence or clear indications, that the use of the RTRBIS concerned is necessary for, and proportionate to, achieving one of the objectives specified in Article 5(1)(h), which need to be identified in the request.⁵¹⁰ Furthermore, the use of such AI system needs to remain limited to what is **strictly necessary** concerning the period of time, as well as the geographic and personal scope. In deciding on the request, that authority shall consider the elements referred to in Article 5(2). If the prior authorization is rejected, the use of the AI system needs to be stopped with immediate effect and all the data, as well as the results and outputs of that use must be immediately discarded and deleted. Recital 12 lists examples of outputs, namely predictions, content, recommendations, or decisions.⁵¹¹

Hidden away in Article 5(3) is the rule that **no decision that produces an adverse legal effect** on a person may be taken based solely on the output of the ‘real-time’ remote biometric identification system.⁵¹² This rule is not about national authorization procedures but rather a general limitation imposed on remote real-time identification systems that would have been better fitted as a third sentence to Article 5(1)(h). This rule needs to be understood also in the context of the qualified prohibition of automated individual decision-making in Article 11(1) LED⁵¹³, which echoes the rule in Article 22(1) GDPR.⁵¹⁴ Considering that the RTRBIS

⁵⁰⁶ Commission Guidelines on prohibited artificial intelligence practices (n 14) 123. See also Hacker (n 19).

⁵⁰⁷ Commission Guidelines on prohibited artificial intelligence practices (n 14) 123. See further Case C-746/18 *Prokuratuur* EU:C:2021:152, para 54; Case C-530/16 *Commission v Poland* EU:C:2018:430, para 67.

⁵⁰⁸ Article 5(3) AIA and Commission Guidelines on prohibited artificial intelligence practices (n 14) 122.

⁵⁰⁹ Article 52(1) CFR.

⁵¹⁰ The CJEU has interpreted the notion of independence of authorities on multiple occasions. See further Case C-746/18 *Prokuratuur* para 54; Case C-530/16 *Commission v Poland* para 67.

⁵¹¹ For an interpretation, see Commission, ‘Guidelines on the definition of an artificial intelligence system established by Regulation (EU) 2024/1689 (AI Act)’ C(2025) 924 final, 10–11. See also analysis of Article 3(1) AIA.

⁵¹² Article 5(3) AIA.

⁵¹³ Article 11(1) LED reads as follows: ‘Member States shall provide for a decision based solely on automated processing, including profiling, which produces an adverse legal effect concerning the data subject or significantly affects him or her, to be prohibited unless authorised by Union or Member State law to which the controller is subject and which provides appropriate safeguards for the rights and freedoms of the data subject, at least the right to obtain human intervention on the part of the controller.’

⁵¹⁴ For a commentary, see Thilo Gottschalk, ‘Automated Individual Decision-Making’ in Eleni Kosta and Franziska Boehm (eds), *The EU Law Enforcement Directive (LED)* (1st edn, OUP 2024).

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also qualify as high-risk AI systems, the obligations in Chapter III also apply. Article 14(5) provides that for biometric identification systems subject to Point 1(a) of Annex III (which includes real-time biometric identification systems but also those that are not real-time), the human oversight measures governed by Article 14(3) need to ensure that no action or decision is taken by the deployer on the basis of the identification resulting from the system, unless that identification has been separately verified and confirmed by at least two natural persons with the necessary competence, training, and authority.⁵¹⁵ Notably, however, EU or national law can provide that this requirement is disproportionate if the AI system is used for purposes of law enforcement, migration, border control, or asylum.⁵¹⁶ Member States may choose to add such limitations to the domestic laws adopted in line with Article 5(5).

- 3.179** Article 5(3) foresees that in a **duly justified situation of urgency**, the use of a RTRBIS can be commenced without an authorization if it is requested without undue delay, at the latest within 24 hours. **Urgency** is a situation where the need to use the systems concerned is such as to make it effectively and objectively impossible to obtain an authorization before commencing the use of the AI system.⁵¹⁷ This rule echoes the general derogation from the conformity assessment procedure that a duly justified situation of urgency for exceptional reasons of public security, or in the case of specific, substantial, and imminent threat to the life or physical safety of natural persons, law enforcement authorities, or civil protection authorities may put a specific high-risk AI system into service without the authorization referred to in Article 46(1).⁵¹⁸ If an AI system is used in a situation of urgency prior to having obtained authorization, then the use of the AI system should ‘be restricted to the absolute minimum necessary and should be subject to appropriate safeguards and conditions, as determined in national law and specified in the context of each individual urgent use case by the law enforcement authority itself’.⁵¹⁹ The subsequent request for authorization must outline the reasons why it was not possible to wait for authorization and explain why the system had to be used immediately due to the urgency of the situation.⁵²⁰

XIII. Article 5(4): Notification to the Relevant Market Surveillance Authority and the National Data Protection Authority

- 3.180** Article 5(4) provides that, without prejudice to Article 5(3), each use of a ‘real-time’ remote biometric identification system in publicly accessible spaces for law enforcement purposes has to be notified to the relevant market surveillance authority and the national data protection authority, in accordance with the national rules that Member States are required to establish under Article 5(5). The notification has to, as a minimum, contain the information specified under Article 5(6), and may not include sensitive operational data. Sensitive operational data is defined by Article 3(38) as operational data related to activities of prevention, detection, investigation, or prosecution of criminal offences, the disclosure of which could jeopardize the integrity of criminal proceedings.⁵²¹

⁵¹⁵ Article 14(5) AIA.

⁵¹⁶ Ibid.

⁵¹⁷ Recital 35 AIA.

⁵¹⁸ Article 46(2) AIA.

⁵¹⁹ Recital 35 AIA.

⁵²⁰ Ibid.

⁵²¹ Article 3(38) AIA.

The market surveillance authorities are national authorities carrying out the activities and taking the measures pursuant to the Market Surveillance Regulation⁵²² In accordance with Article 70(1), each Member State has to establish or designate at least one market surveillance authority,⁵²³ which has to cooperate and exchange in the context of one of the two standing subgroups to be established under the Artificial Intelligence Board.⁵²⁴ In addition, national data protection authorities also need to be notified. This is since biometric identification systems, by definition, process personal data, as defined by Article 4(1) GDPR.⁵²⁵

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XIV. Article 5(5): Full or Partial Authorization by a Member State

Although Article 5(5) appears later in the regulatory framework, it establishes the foundational principle that real-time remote biometric identification systems for law enforcement purposes may only be authorized pursuant to Member State law. Article 5(5) provides that Member States can fully or partially authorize the use of real-time remote biometric identification systems for the purposes of law enforcement within the limits set out by Article 5(1) (h), as well as 5(2) and 5(3). Member States must lay down in their national law the necessary detailed rules for the request, issuance and exercise of, as well as supervision and reporting relating to, the authorizations referred to in Article 5(3). Those rules must also specify in respect of which of the objectives listed in Article 5(1)(h), including which of the criminal offences referred to in Article 5(1)(h)(iii), the competent authorities may be authorized to use those systems for the purposes of law enforcement. Member States must notify those rules to the Commission at the latest 30 days following the adoption thereof. Member States may introduce, in accordance with EU law, more restrictive laws on the use of remote biometric identification systems. Such use shall only be possible where and in as far as the Member State concerned has decided to expressly provide for the possibility to authorize such use in its detailed rules of national law.⁵²⁶ Consequently, Member States are also free not to provide for such a possibility at all or to only provide for such a possibility in respect of some of the objectives capable of justifying authorized use identified in the AI Act.⁵²⁷ Where Member States adopt related rules, they should be notified to the Commission within 30 days of their adoption.⁵²⁸

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These rules underline that Article 5(2) functions as a floor for protection.⁵²⁹ Unlike other provisions, Article 5(2) establishes a minimum threshold of protection rather than

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⁵²² See further Article 3(26) AIA and MSR.

⁵²³ For an overview of these developments, see 'Responsible Authorities for the Enforcement of the AI Act on National Level' (*gdpr.blog.hu*, 2 October 2024) <https://gdpr.blog.hu/2024/10/02/responsible_authorities_for_the_enforcement_of_the_ai_act_on_national_level> accessed 24 December 2024.

⁵²⁴ Article 65(6) AIA.

⁵²⁵ Article 4(1) GDPR defines personal data as 'any information relating to an identified or identifiable natural person ('data subject'); an identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person.'

⁵²⁶ Recital 37 AIA.

⁵²⁷ *Ibid.*

⁵²⁸ *Ibid.*

⁵²⁹ On floors, and ceilings, for fundamental rights in EU law see further Federico Fabbrini, *Fundamental Rights in Europe* (OUP, 2014).

a maximum limit. It permits Member States, in alignment with Union law, to implement stricter regulations on the use of remote biometric identification systems.

XV. Article 5(6): Annual Reports by National Market Surveillance Authorities and National Data Protection Authorities

- 3.184** Article 5(6) provides that national market surveillance authorities and the national data protection authorities that have been notified of the use of ‘real-time’ remote biometric identification systems in publicly accessible spaces for law enforcement purposes pursuant to Article 5(4) shall submit to the European Commission annual reports on such use.
- 3.185** For that purpose, the Commission must provide Member States and national market surveillance and data protection authorities with a template, including information on the number of the decisions taken by competent judicial authorities or an independent administrative authority whose decision is binding upon requests for authorizations in accordance with Article 5(3) and their result.

XVI. Article 5(7): Annual Reports by the European Commission

- 3.186** Article 5(7) requires the European Commission to publish annual reports on the use of real-time remote biometric identification systems in publicly accessible spaces for law enforcement purposes, based on aggregated data in Member States on the basis of the annual reports referred to in Article 5(6). Those annual reports shall not include sensitive operational data of the related law enforcement activities. Sensitive operational data is defined by Article 3(38) as operational data related to activities of prevention, detection, investigation, or prosecution of criminal offences, the disclosure of which could jeopardize the integrity of criminal proceedings.⁵³⁰

XVII. Article 5(8): Practices Otherwise Contrary to EU Law

- 3.187** Article 5(8) provides that Article 5 does not affect the prohibitions that apply where an AI practice infringes other Union law. The exception to this rule is that the AI Act framework on biometric surveillance by law enforcement serves as a *lex specialis* to Article 10 LED.⁵³¹

⁵³⁰ Article 3(38) AIA.

⁵³¹ Recital 38 AIA. For an analysis, see further above.